Physics - Class 11

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Chapter 1 - Measurements

Exercise 1 - Questions

1.1 Name several repetitive phenomena occurring in nature which could serve as reasonable time standards.

The rotation of the Earth on its axis (day and night), the revolution of the Earth around the Sun (a year), the phases of the Moon (a lunar month), the tides, the swinging of a pendulum, the vibration of a quartz crystal, the beating of the heart and the sunrise and sunset are all repetitive phenomena. Any phenomenon that repeats itself after a fixed, equal interval of time can serve as a time standard.

1.2 Give the drawbacks to use the period of a pendulum as a time standard.

The period of a pendulum depends on the value of g, which changes from place to place, and on the length of the pendulum, which changes with temperature. Friction at the support and air resistance also change the motion, and the period is not exactly the same for large swings. A pendulum is therefore not perfectly reproducible, is hard to carry from place to place, and its small errors add up over a long time.

1.3 Why do we find it useful to have two units for the amount of substance, the kilogram and the mole?

The kilogram measures mass, that is, how much matter a body contains. The mole measures the amount of substance, that is, how many particles (atoms or molecules) it contains. One mole contains 6.02 x 10^23 particles. Equal masses of

different substances contain different numbers of particles, so in chemistry, where reactions depend on the number of particles, the mole is needed in addition to the kilogram.

1.4 Three students measured the length of a needle with a scale on which minimum division is 1 mm and recorded as (i) 0.2145 m (ii) 0.21 m (iii) 0.214 m. Which record is correct and why?

Record (iii) 0.214 m is correct. A scale with a minimum division of 1 mm = 0.001 m can be read reliably only up to three places of decimals in metres. Record (i) 0.2145 m claims a precision of 0.1 mm, which the scale cannot give. Record (ii) 0.21 m is less precise than the scale allows, because it drops a figure the scale can actually measure.

1.5 An old saying is that 'A chain is only as strong as its weakest link'. What analogous statement can you make regarding experimental data used in a computation?

A result obtained from experimental data is only as accurate as the least accurate measurement used in it. In a computation, the final answer cannot have more significant figures, or more decimal places in addition and subtraction, than the least precise measurement that went into it.

1.6 The period of simple pendulum is measured by a stop watch. What type of errors are possible in the time period?

Both systematic and random errors are possible. A systematic error comes from the stop watch itself, for example a zero error or a wrong calibration, which affects every reading in the same way. Random errors come from the observer, mainly the human reaction time in starting and stopping the watch, which is different at each trial, and from small variations in counting the vibrations.

1.7 Does a dimensional analysis give any information on constant of proportionality that may appear in an algebraic expression? Explain.

No. Dimensional analysis only fixes the powers of the quantities in a relation, because a constant of proportionality is dimensionless and does not change the dimensions of either side. The numerical value of the constant, for example 2 pi in T = 2 pi sqrt(l/g), cannot be found by dimensional analysis; it must be found by experiment or by a full theory.

1.8 Write the dimensions of (i) Pressure (ii) Density.

(i) Pressure = force / area. Dimensions = [M L T^-2] / [L^2] = [M L^-1 T^-2]. (ii) Density = mass / volume. Dimensions = [M] / [L^3] = [M L^-3].

1.9 The wavelength lambda of a wave depends on the speed v of the wave and its frequency f. Knowing that [lambda] = [L], [v] = [L T^-1] and [f] = [T^-1], decide which of the following is correct: f = v lambda or f = v / lambda.

Check f = v lambda: dimensions of v lambda = [L T^-1] x [L] = [L^2 T^-1], which is not [T^-1]. Check f = v / lambda: dimensions of v / lambda = [L T^-1] / [L] = [T^-1], which equals [f]. Therefore f = v / lambda is correct.

Exercise 1 - Numerical Problems

1.1 A light year is the distance light travels in one year. How many metres are there in one light year? (speed of light = 3.0 x 10^8 ms^-1).

Step 1 - Time in one year: t = 365 days x 24 hours/day x 3600 s/hour.

Step 2 - t = 365 x 24 x 3600 s = 3.1536 x 10^7 s.

Step 3 - Distance = speed x time: d = (3.0 x 10^8 ms^-1) x (3.1536 x 10^7 s).

Step 4 - d = 9.4608 x 10^15 m, which to two significant figures is 9.5 x 10^15 m.

Answer: 9.5 x 10^15 m (as printed in the book).

1.2 (a) How many seconds are there in 1 year? (b) How many nanoseconds in 1 year? (c) How many years in 1 second?

Step 1 - (a) Seconds in 1 year = 365 x 24 x 3600 = 3.1536 x 10^7 s.

Step 2 - (b) 1 s = 10^9 ns, so nanoseconds in 1 year = 3.1536 x 10^7 x 10^9 = 3.1536 x 10^16 ns.

Step 3 - (c) Years in 1 second = 1 / (3.1536 x 10^7) year = 3.17 x 10^-8 year.

Answer: (a) 3.1536 x 10^7 s, (b) 3.1536 x 10^16 ns, (c) 3.17 x 10^-8 year. Note: the book prints (c) as 3.1 x 10^-8 year, a rougher rounding; the value from 365 days is 3.17 x 10^-8 year.

1.3 The length and width of a rectangular plate are measured to be 15.3 cm and 12.80 cm, respectively. Find the area of the plate.

Step 1 - Area = length x width = 15.3 cm x 12.80 cm.

Step 2 - Area = 195.84 cm^2.

Step 3 - The least precise measurement, 15.3 cm, has three significant figures, so the result is rounded to three significant figures.

Answer: 196 cm^2 (as printed in the book).

1.4 Add the following masses given in kg up to appropriate precision: 2.189, 0.089, 11.8 and 5.32.

Step 1 - Add the masses: 2.189 + 0.089 + 11.8 + 5.32 = 19.398 kg.

Step 2 - In addition, the result is rounded to the decimal place of the least precise measurement. The least precise value is 11.8 kg, which is known only to one place of decimals.

Step 3 - 19.398 kg rounded to one place of decimals is 19.4 kg.

Answer: 19.4 kg (as printed in the book).

1.5 Find the value of 'g' and its uncertainty using T = 2 pi sqrt(l/g) from the following measurements: length of simple pendulum l = 100 cm; time for 20 vibrations = 40.2 s; length measured by a metre scale of accuracy up to 1 mm and time by stop watch of accuracy up to 0.1 s.

Step 1 - Time period for one vibration: T = 40.2 s / 20 = 2.01 s.

Step 2 - Length in metres: l = 100 cm = 1.00 m.

Step 3 - From T = 2 pi sqrt(l/g), squaring both sides gives g = 4 pi^2 l / T^2.

Step 4 - With pi = 3.14: g = 4 x (3.14)^2 x 1.00 / (2.01)^2 = 39.4384 / 4.0401 = 9.76 ms^-2.

Step 5 - Uncertainty in length: delta l = 1 mm = 0.001 m, so delta l / l = 0.001 / 1.00 = 0.001.

Step 6 - Uncertainty in one period: delta T = 0.1 s / 20 = 0.005 s, so delta T / T = 0.005 / 2.01 = 0.00249.

Step 7 - Fractional uncertainty in g: delta g / g = delta l / l + 2 x (delta T / T) = 0.001 + 2 x 0.00249 = 0.00597.

Step 8 - delta g = 9.76 x 0.00597 = 0.058 ms^-2, about 0.06 ms^-2.

Answer: g = 9.76 +- 0.06 ms^-2 (as printed in the book).

1.6 What are the dimensions and units of gravitational constant G in the formula F = G m1 m2 / r^2?

Step 1 - Rearrange the formula: G = F r^2 / (m1 m2).

Step 2 - Dimensions: [G] = [M L T^-2] x [L^2] / [M^2].

Step 3 - [G] = [M^-1 L^3 T^-2].

Step 4 - Unit: from F = G m1 m2 / r^2, the unit of G is N m^2 kg^-2.

Answer: [M^-1 L^3 T^-2], N m^2 kg^-2 (as printed in the book).

1.7 Show that the expression vf = vi + at is dimensionally correct, where vi is the velocity at t = 0, a is acceleration and vf is the velocity at time t.

Step 1 - Dimensions of the left hand side: [vf] = [L T^-1].

Step 2 - Dimensions of vi: [vi] = [L T^-1].

Step 3 - Dimensions of at: [a] x [t] = [L T^-2] x [T] = [L T^-1].

Step 4 - Both terms on the right hand side have the same dimensions, [L T^-1], and they equal the dimensions of the left hand side.

Answer: the expression is dimensionally correct (this shows consistency of dimensions, not that the equation is fully proved).

1.8 The speed v of sound waves through a medium may be assumed to depend on (a) the density rho of the medium and (b) its modulus of elasticity E which is the ratio of stress to strain. Deduce by the method of dimensions, the formula for the speed of sound.

Step 1 - Assume v = Constant x rho^a x E^b, where a and b are to be found.

Step 2 - Dimensions: [v] = [L T^-1], [rho] = [M L^-3], [E] = [M L^-1 T^-2] (the same dimensions as pressure).

Step 3 - [L T^-1] = ([M L^-3])^a x ([M L^-1 T^-2])^b = [M^(a+b) L^(-3a-b) T^(-2b)].

Step 4 - Equate the powers of M: a + b = 0.

Step 5 - Equate the powers of T: -2b = -1, so b = 1/2.

Step 6 - From a + b = 0, a = -1/2. Check the powers of L: -3a - b = -3(-1/2) - 1/2 = 1, as required.

Step 7 - Therefore v = Constant x rho^(-1/2) x E^(1/2) = Constant x sqrt(E / rho).

Answer: v = Constant sqrt(E / rho) (as printed in the book).

1.9 Show that the famous 'Einstein equation' E = mc^2 is dimensionally consistent.

Step 1 - Dimensions of energy E: [E] = [M L^2 T^-2].

Step 2 - Dimensions of mc^2: [m] x [c]^2 = [M] x ([L T^-1])^2 = [M L^2 T^-2].

Step 3 - Both sides have the same dimensions, [M L^2 T^-2].

Answer: E = mc^2 is dimensionally consistent.

1.10 Suppose, we are told that the acceleration of a particle moving in a circle of radius r with uniform speed v is proportional to some power of r, say r^n, and some power of v, say v^m, determine the powers of r and v.

Step 1 - Write a = Constant x r^n x v^m.

Step 2 - Dimensions: [a] = [L T^-2], [r]^n = [L^n], [v]^m = ([L T^-1])^m = [L^m T^-m].

Step 3 - So [L T^-2] = [L^(n+m) T^-m].

Step 4 - Equate the powers of T: -m = -2, so m = 2.

Step 5 - Equate the powers of L: n + m = 1, so n = 1 - 2 = -1.

Answer: n = -1, m = 2 (as printed in the book), that is, a is proportional to v^2 / r.

Chapter 2 - Vectors and Equilibrium

Exercise 2 - Questions

2.1 Define the terms (i) unit vector (ii) Position vector and (iii) Components of a vector.

(i) A unit vector is a vector of magnitude one. It only shows a direction; for example, i, j and k are unit vectors along the x, y and z axes. (ii) A position vector is the vector drawn from the origin of a coordinate system to a point; it gives the position of that point. (iii) The components of a vector are its projections on the coordinate axes. For a vector A in three dimensions, Ax, Ay and Az are its rectangular components, and A = Ax i + Ay j + Az k.

2.2 The vector sum of three vectors gives a zero resultant. What can be the orientation of the vectors?

When the three vectors are placed head to tail, they form a closed triangle. This means the three vectors are coplanar and are arranged so that the head of the last vector meets the tail of the first, giving a zero resultant.

2.3 Vector A lies in the xy plane. For what orientation will both of its rectangular components be negative? For what orientation will its components have opposite signs?

Both components are negative when A lies in the third quadrant, that is, when it makes an angle between 180 degrees and 270 degrees with the positive x-axis. The components have opposite signs when A lies in the second quadrant (Ax negative, Ay positive) or in the fourth quadrant (Ax positive, Ay negative).

2.4 If one of the rectangular components of a vector is not zero, can its magnitude be zero? Explain.

No. The magnitude of a vector is the square root of the sum of the squares of its

components, so it is at least as large as any one component. If even one component is not zero, the magnitude is greater than zero.

2.5 Can a vector have a component greater than the vector's magnitude?

No. A component is the magnitude multiplied by the cosine of an angle, and the cosine can never be greater than one in size. So no component can be greater than the magnitude of the vector itself.

2.6 Can the magnitude of a vector have a negative value?

No. The magnitude of a vector is a length, and a length is always zero or positive. It can never be negative.

2.7 If A + B = 0, what can you say about the components of the two vectors?

Then B = -A. The two vectors are equal in magnitude and opposite in direction, so their corresponding components are equal in size but opposite in sign: Bx = -Ax, By = -Ay and Bz = -Az.

2.8 Under what circumstances would a vector have components that are equal in magnitude?

When the vector makes equal angles with the axes, so that its direction cosines are equal in size. In two dimensions this happens when the vector makes an angle of 45 degrees with the x-axis, giving Ax = Ay.

2.9 Is it possible to add a vector quantity to a scalar quantity? Explain.

No. Only quantities of the same kind can be added. A vector has magnitude and direction, while a scalar has magnitude only, so adding them has no meaning in physics.

2.10 Can you add zero to a null vector?

A scalar zero cannot be added to a vector. But the null vector itself can be added to a null vector, and the result is still the null vector, because a null vector has zero magnitude.

2.11 Two vectors have unequal magnitudes. Can their sum be zero? Explain.

No. For the sum to be zero the two vectors must be equal in magnitude and opposite in direction, so that they cancel completely. With unequal magnitudes the smallest possible resultant is the difference of the two magnitudes, which is not zero.

2.12 Show that the sum and difference of two perpendicular vectors of equal lengths are also perpendicular and of the same length.

Let the vectors be A and B, perpendicular, each of length L. Their sum is A + B and their difference is A - B. Their dot product is (A + B).(A - B) = |A|^2 - |B|^2 = L^2 - L^2 = 0, so they are perpendicular. Their lengths are |A + B| = sqrt(L^2 + L^2) = L sqrt(2) and |A - B| = sqrt(L^2 + L^2) = L sqrt(2), so they are equal.

2.13 How would the two vectors of the same magnitude have to be oriented, if they were to be combined to give a resultant equal to a vector of the same magnitude?

They must be at an angle of 120 degrees to each other. If each vector has magnitude A, the resultant R satisfies R^2 = A^2 + A^2 + 2 A^2 cos(theta). Setting R = A gives cos(theta) = -1/2, so theta = 120 degrees.

2.14 The two vectors to be combined have magnitudes 60 N and 35 N. Pick the correct answer from those given below and tell why is it the only one of the three that is correct. (i) 100 N (ii) 70 N (iii) 20 N

The correct answer is (ii) 70 N. The magnitude of the resultant of two vectors must lie between the difference and the sum of their magnitudes, that is, between 60 - 35 = 25 N and 60 + 35 = 95 N. Only 70 N lies in this range; 100 N is greater than the sum and 20 N is less than the difference, so both are impossible.

2.15 Suppose the sides of a closed polygon represent vector arranged head to tail. What is the sum of these vectors?

The sum is zero (the null vector). Going around a closed polygon head to tail brings you back to the starting point, so there is no net displacement.

2.16 Identify the correct answer. (i) Two ships X and Y are travelling in different

directions at equal speeds. The actual direction of motion of X is due north but to an observer on Y, the apparent direction of motion of X is north-east. The actual direction of motion of Y as observed from the shore will be (A) East (B) West (C) south-east (D) south-west. (ii) A horizontal force F is applied to a small object P of mass m at rest on a smooth plane inclined at an angle theta to the horizontal as shown in Fig. 2.21. The magnitude of the resultant force acting up and along the surface of the plane, on the object is (a) F cos(theta) - m g sin(theta) (b) F sin(theta) - m g cos(theta) (c) F cos(theta) + m g cos(theta) (d) F sin(theta) + m g sin(theta) (e) m g tan(theta).

(i) The correct answer is (B) West. The apparent velocity of X as seen from Y is the velocity of X minus the velocity of Y. X moves north and Y moves west at the same speed, so the apparent velocity points north-east, as given. (ii) The correct answer is (a) F cos(theta) - m g sin(theta). The component of the horizontal force F up the plane is F cos(theta), and the component of the weight down the plane is m g sin(theta). Their difference is the net force up the plane.

2.17 If all the components of the vectors, A1 and A2 were reversed, how would this alter A1 x A2?

It would not alter it at all. Reversing all components changes A1 to -A1 and A2 to -A2, and (-A1) x (-A2) = A1 x A2, because the two minus signs cancel.

2.18 Name the three different conditions that could make A1 x A2 = 0.

The cross product is zero if (1) A1 is the null vector, (2) A2 is the null vector, or (3) A1 and A2 are parallel or antiparallel to each other, so that the angle between them is 0 degrees or 180 degrees and sin(theta) = 0.

2.19 Identify true or false statements and explain the reason. (a) A body in equilibrium implies that it is not moving nor rotating. (b) If coplanar forces acting on a body form a closed polygon, then the body is said to be in equilibrium.

(a) False. Equilibrium means zero acceleration, not zero velocity. A body in

equilibrium may be at rest, or it may move with a uniform velocity and rotate uniformly, because then its translational and angular accelerations are both zero.

(b) True. If coplanar forces placed head to tail form a closed polygon, their vector sum is zero, which is the condition for translational equilibrium.

2.20 A picture is suspended from a wall by two strings. Show by diagram the configuration of the strings for which the tension in the strings will be minimum.

The tension is minimum when the two strings are vertical and parallel to each other. In that case each string carries half the weight of the picture, T = W/2. If the strings are spread apart, each string must also balance the sideways pull of the other, so the tension grows; it becomes very large when the strings are nearly horizontal.

Drawing instruction: draw the picture hanging from two separate, parallel vertical strings, each carrying half the weight.

2.21 Can a body rotate about its centre of gravity under the action of its weight?

No. The weight of a body acts at its centre of gravity, so the line of action of the weight passes through the centre of gravity itself. The moment arm about the centre of gravity is therefore zero, and the weight produces no torque about that point.

Exercise 2 - Numerical Problems

2.1 Suppose, in a rectangular coordinate system, a vector A has its tail at the point P (-2, -3) and its tip at Q (3, 9). Determine the distance between these two points.

Step 1 - Change in x: dx = 3 - (-2) = 5.

Step 2 - Change in y: dy = 9 - (-3) = 12.

Step 3 - Distance = sqrt(dx^2 + dy^2) = sqrt(25 + 144) = sqrt(169) = 13.

Answer: 13 units (as printed in the book).

2.2 A certain corner of a room is selected as the origin of a rectangular coordinate system. If an insect is sitting on an adjacent wall at a point having coordinates (2, 1), where the units are in metres, what is the distance of the insect from this corner of the room?

Step 1 - Distance = sqrt(2^2 + 1^2) = sqrt(5).

Step 2 - sqrt(5) = 2.236 m, about 2.2 m.

Answer: 2.2 m (as printed in the book).

2.3 What is the unit vector in the direction of the vector A = 4 i + 3 j?

Step 1 - Magnitude of A: |A| = sqrt(4^2 + 3^2) = sqrt(25) = 5.

Step 2 - Unit vector = A / |A| = (4 i + 3 j) / 5.

Answer: (4 i + 3 j) / 5 (as printed in the book).

2.4 Two particles are located at r1 = 3 i + 7 j and r2 = -2 i + 3 j respectively. Find both the magnitude of the vector r2 - r1 and its orientation with respect to the x-axis.

Step 1 - r2 - r1 = (-2 - 3) i + (3 - 7) j = -5 i - 4 j.

Step 2 - Magnitude = sqrt((-5)^2 + (-4)^2) = sqrt(41) = 6.4.

Step 3 - The vector lies in the third quadrant. tan(phi) = (-4)/(-5) = 0.8, so phi =

38.7 degrees.

Step 4 - Angle with the positive x-axis = 180 degrees + 38.7 degrees = 218.7 degrees, about 219 degrees.

Answer: 6.4, 219 degrees (as printed in the book).

2.5 If a vector B is added to vector A, the result is 6 i + j. If B is subtracted from A, the result is -4 i + 7 j. What is the magnitude of vector A?

Step 1 - A + B = 6 i + j and A - B = -4 i + 7 j.

Step 2 - Add the two equations: 2A = (6 - 4) i + (1 + 7) j = 2 i + 8 j.

Step 3 - So A = i + 4 j.

Step 4 - Magnitude of A = sqrt(1^2 + 4^2) = sqrt(17) = 4.1.

Answer: 4.1 (as printed in the book).

2.6 Given that A = 2 i + 3 j and B = 3 i - 4 j, find the magnitude and angle of (a) C = A + B, and (b) D = 3A - 2B.

Step 1 - (a) C = A + B = (2 + 3) i + (3 - 4) j = 5 i - j.

Step 2 - Magnitude of C = sqrt(25 + 1) = sqrt(26) = 5.1.

Step 3 - Angle of C: tan(theta) = -1/5, so theta = -11.3 degrees, which is 360 - 11.3 = 349 degrees from the positive x-axis.

Step 4 - (b) D = 3A - 2B = (6 - 6) i + (9 + 8) j = 17 j.

Step 5 - Magnitude of D = 17, and its angle with the positive x-axis is 90 degrees.

Answer: 5.1, 349 degrees; 17, 90 degrees (as printed in the book).

2.7 Find the angle between the two vectors, A = 5 i + j and B = 2 i + 4 j.

Step 1 - Dot product: A.B = (5 x 2) + (1 x 4) = 14.

Step 2 - |A| = sqrt(25 + 1) = sqrt(26) = 5.10.

Step 3 - |B| = sqrt(4 + 16) = sqrt(20) = 4.47.

Step 4 - cos(theta) = (A.B) / (|A| |B|) = 14 / (5.10 x 4.47) = 14 / 22.80 = 0.614.

Step 5 - theta = cos^-1(0.614) = 52 degrees.

Answer: 52 degrees (as printed in the book).

2.8 Find the work done when the point of application of the force 3 i + 2 j moves in a straight line from the point (2, -1) to the point (6, 4).

Step 1 - Displacement = (6 - 2) i + (4 - (-1)) j = 4 i + 5 j.

Step 2 - Work = F.d = (3 x 4) + (2 x 5) = 12 + 10 = 22.

Answer: 22 units (as printed in the book).

2.9 Show that the three vectors i + j + k, 2 i - 3 j + k and 4 i + j - 5 k are mutually perpendicular.

Step 1 - Dot product of the first and second: (1 x 2) + (1 x -3) + (1 x 1) = 2 - 3 + 1 = 0.

Step 2 - Dot product of the first and third: (1 x 4) + (1 x 1) + (1 x -5) = 4 + 1 - 5 = 0.

Step 3 - Dot product of the second and third: (2 x 4) + (-3 x 1) + (1 x -5) = 8 - 3 - 5 = 0.

Step 4 - All three dot products are zero, so each pair is perpendicular.

Answer: the three vectors are mutually perpendicular.

2.10 Given that A = i - 2 j + 3 k and B = 3 i - 4 k, find the projection of A on B.

Step 1 - Dot product: A.B = (1 x 3) + (-2 x 0) + (3 x -4) = 3 - 12 = -9.

Step 2 - Magnitude of B: |B| = sqrt(9 + 16) = 5.

Step 3 - Scalar projection of A on B = (A.B) / |B| = -9 / 5.

Answer: -9/5 (as printed in the book). The negative sign means the projection points opposite to B.

2.11 Vectors A, B and C are 4 units north, 3 units west and 8 units east, respectively. Describe carefully (a) A x B (b) A x C (c) B x C.

Step 1 - Take north as j and east as i. Then A = 4 j, B = -3 i, C = 8 i.

Step 2 - (a) A x B = 4 j x (-3 i) = -12 (j x i) = -12 (-k) = 12 k. Magnitude 12 units, direction vertically up.

Step 3 - (b) A x C = 4 j x 8 i = 32 (j x i) = -32 k. Magnitude 32 units, direction vertically down.

Step 4 - (c) B x C = (-3 i) x (8 i) = 0, because the vectors are parallel (both along the east-west line).

Answer: (a) 12 units vertically up (b) 32 units vertically down (c) Zero (as printed in the book).

2.12 The torque or turning effect of force about a given point is given by r x F where r is the vector from the given point to the point of application of F. Consider a force F = -3 i + j + 5 k (newton) acting on the point 7 i + 3 j + k (m). What is the torque in N m about the origin?

Step 1 - Torque = r x F, with r = 7 i + 3 j + k and F = -3 i + j + 5 k.

Step 2 - i-component: (3 x 5) - (1 x 1) = 15 - 1 = 14.

Step 3 - j-component: -[(7 x 5) - (1 x -3)] = -(35 + 3) = -38.

Step 4 - k-component: (7 x 1) - (3 x -3) = 7 + 9 = 16.

Answer: 14 i - 38 j + 16 k N m (as printed in the book).

2.13 The line of action of force, F = i - 2 j, passes through a point whose position vector is (-j + k). Find (a) the moment of F about the origin, (b) the moment of F about the point of which the position vector is i + k.

Step 1 - (a) Moment about the origin = r x F with r = -j + k and F = i - 2 j.

Step 2 - i-component: (-1 x 0) - (1 x -2) = 2.

Step 3 - j-component: -[(0 x 0) - (1 x 1)] = 1.

Step 4 - k-component: (0 x -2) - (-1 x 1) = 1. So the moment about the origin is 2 i + j + k.

Step 5 - (b) Vector from the new point to the point on the line of action: (-j + k) - (i + k) = -i - j.

Step 6 - Moment = (-i - j) x (i - 2 j) = (-i) x (-2 j) + (-j) x (i) = 2 k + k = 3 k.

Answer: (a) 2 i + j + k, (b) 3 k (as printed in the book).

2.14 The magnitude of dot and cross products of two vectors are 6 sqrt(3) and 6 respectively. Find the angle between the vectors.

Step 1 - |A x B| = A B sin(theta) = 6 and A.B = A B cos(theta) = 6 sqrt(3).

Step 2 - Divide: tan(theta) = |A x B| / (A.B) = 6 / (6 sqrt(3)) = 1 / sqrt(3).

Step 3 - theta = tan^-1(1 / sqrt(3)) = 30 degrees.

Answer: 30 degrees (as printed in the book).

2.15 A load of 10.0 N is suspended from a clothes line. This distorts the line so that it makes an angle of 15 degrees with the horizontal at each end. Find the tension in the clothes line.

Step 1 - At the load, the two halves of the line pull up and outwards, each with tension T at 15 degrees to the horizontal.

Step 2 - Vertical equilibrium: the upward components balance the weight: 2 T sin(15 degrees) = 10.0 N.

Step 3 - sin(15 degrees) = 0.2588, so 2 T x 0.2588 = 10.0.

Step 4 - T = 10.0 / 0.5176 = 19.3 N.

Answer: 19.3 N (as printed in the book).

2.16 A tractor of weight 15,000 N crosses a single span bridge of weight 8000 N and of length 21.0 m. The bridge span is supported half a metre from either end. The tractor's front wheels take 1/3 of the total weight of the tractor, and the rear wheels are 3 m behind the front wheels. Calculate the force on the bridge supports when the rear wheels are at the middle of the bridge span.

Step 1 - Supports are 0.5 m from each end, so the distance between the supports is 21.0 - 0.5 - 0.5 = 20.0 m. Call the left support A and the right support B.

Step 2 - Rear wheels at the middle of the span: 10.5 m from the left end, carrying 2/3 x 15,000 = 10,000 N. Front wheels 3 m ahead: 13.5 m from the left end,

carrying 1/3 x 15,000 = 5,000 N.

Step 3 - Bridge weight 8,000 N acts at its centre, 10.5 m from the left end. Total downward force = 10,000 + 5,000 + 8,000 = 23,000 N.

Step 4 - Take moments about A (0.5 m from the left end). Distances from A: rear wheels 10.0 m, front wheels 13.0 m, bridge centre 10.0 m.

Step 5 - R_B x 20.0 = (10,000 x 10.0) + (5,000 x 13.0) + (8,000 x 10.0) = 100,000 + 65,000 + 80,000 = 245,000.

Step 6 - R_B = 245,000 / 20.0 = 12,250 N = 12.25 kN.

Step 7 - R_A = 23,000 - 12,250 = 10,750 N = 10.75 kN.

Answer: 12.25 kN and 10.75 kN (as printed in the book; the larger force is on the right support, nearer the front wheels).

2.17 A spherical ball of weight 50 N is to be lifted over the step as shown in the Fig. 2.23. Calculate the minimum force needed just to lift it above the floor. (Fig. 2.23: ball radius 20 cm, step height 5 cm, horizontal force F whose line of action is 10 cm above the centre of the ball.)

Step 1 - Take the corner of the step as the pivot. Just as the ball starts to lift, it turns about this corner.

Step 2 - Radius R = 20 cm, step height h = 5 cm. The centre of the ball is R - h = 15 cm above the pivot level.

Step 3 - Horizontal distance from the pivot to the vertical line through the centre: sqrt(R^2 - (R - h)^2) = sqrt(20^2 - 15^2) = sqrt(400 - 225) = sqrt(175) = 13.23 cm.

Step 4 - Moment of the weight about the pivot: 50 N x 13.23 cm (this resists the lift).

Step 5 - The line of action of F is 10 cm above the centre, so its height above the pivot level is 10 + 15 = 25 cm. Moment of F about the pivot: F x 25 cm.

Step 6 - For just lifting, the moments balance: F x 25 = 50 x 13.23, so F = 661.5 / 25 = 26.5 N, about 26 N.

Answer: 26 N (as printed in the book).

2.18 A uniform sphere of weight 10.0 N is held by a string attached to a frictionless wall so that the string makes an angle of 30 degrees with the wall as shown in Fig. 2.24. Find the tension in the string and the force exerted on the sphere by the wall.

Step 1 - Forces on the sphere: weight 10.0 N downwards, tension T along the string, and the normal force N from the wall, horizontal.

Step 2 - The string makes 30 degrees with the wall (the vertical), so the vertical component of T is T cos(30 degrees) and the horizontal component is T sin(30 degrees).

Step 3 - Vertical equilibrium: T cos(30 degrees) = 10.0 N, so T = 10.0 / 0.8660 = 11.55 N, about 11.6 N.

Step 4 - Horizontal equilibrium: N = T sin(30 degrees) = 11.55 x 0.5 = 5.77 N.

Answer: 11.6 N, 5.77 N (as printed in the book).

Chapter 3 - Motion and Force

Exercise 3 - Questions

3.1 What is the difference between uniform and variable velocity? From the explanation of variable velocity, define acceleration. Give SI units of velocity and acceleration.

Uniform velocity means a body covers equal displacements in equal intervals of time, in the same direction. Variable velocity means the velocity changes, either in magnitude or in direction or both. Acceleration is the rate of change of velocity, a = change in velocity / time taken. The SI unit of velocity is metre per second (ms^-1) and of acceleration is metre per second squared (ms^-2).

3.2 An object is thrown vertically upward. Discuss the sign of acceleration due to gravity, relative to velocity, while the object is in air.

Take the upward direction as positive. The acceleration due to gravity is always downwards, so it is always negative, -g, throughout the flight. While the object rises, its velocity is positive and decreasing, so acceleration is opposite to velocity. At the highest point the velocity is zero for an instant, but the acceleration is still -g. While the object falls, its velocity is negative and growing in size, so acceleration is in the same direction as velocity.

3.3 Can the velocity of an object reverse the direction when acceleration is constant? If so, give an example.

Yes. A ball thrown vertically upward is an example: its acceleration is the constant acceleration due to gravity, downwards, yet its velocity is first upward, becomes zero at the top, and then becomes downward. The velocity reverses its direction while the acceleration stays constant.

3.4 Specify the correct statements: (a) An object can have a constant velocity even its speed is changing. (b) An object can have a constant speed even its velocity is changing. (c) An object can have a zero velocity even its acceleration is not zero. (d) An object subjected to a constant acceleration can reverse its velocity.

(a) False. Constant velocity means both magnitude and direction are unchanged, so the speed, which is the magnitude of velocity, cannot change. (b) True. In uniform circular motion the speed stays constant while the direction of velocity changes all the time. (c) True. At the highest point of a vertical throw the velocity is zero for an instant, but the acceleration due to gravity is still acting. (d) True. An object thrown upward under constant acceleration due to gravity rises, stops, and then falls back, so its velocity reverses.

3.5 A man standing on the top of a tower throws a ball straight up with initial velocity vi, and at the same time throws a second ball straight downward with the same speed. Which ball will have larger speed when it strikes the ground? Ignore air friction.

Both balls strike the ground with the same speed. The ball thrown upward returns to the level of the tower top with the same speed vi, now directed downward, which is exactly the starting condition of the ball thrown downward. From there both fall through the same height under the same gravity, so both reach the ground with speed sqrt(vi^2 + 2 g h). Air friction is ignored.

3.6 Explain the circumstances in which the velocity v and acceleration a of a car are (i) Parallel (ii) Anti-parallel (iii) Perpendicular to one another (iv) v is zero but a is not zero (v) a is zero but v is not zero.

(i) Parallel: velocity and acceleration point in the same direction, for example a car speeding up on a straight road. (ii) Anti-parallel: acceleration points opposite to velocity, for example a car braking on a straight road. (iii) Perpendicular to one another: acceleration is at right angles to velocity, for example a car going round a bend at constant speed, where the acceleration is centripetal. (iv) v is zero but a is not zero: at the instant a ball thrown upward reaches its highest point, or at the instant a car starts from rest. (v) a is zero but v is not zero: a car moving with uniform velocity on a straight road, with no net force acting.

3.7 Motion with constant velocity is a special case of motion with constant acceleration. Is this statement true? Discuss.

True. Motion with constant velocity is motion with a constant acceleration of zero. Acceleration is the rate of change of velocity; if the velocity does not change, its rate of change is zero, which is a constant value. All the equations of uniformly accelerated motion then reduce to the uniform-motion results, for example distance = velocity x time.

3.8 Find the change in momentum for an object subjected to a given force for a given time and state law of motion in terms of momentum.

The change in momentum is the product of the force and the time for which it acts: change in momentum = F x t. This is the impulse of the force. Newton's second law of motion in terms of momentum states that the force acting on a body is equal to the rate of change of its momentum: F = change in momentum / time taken.

3.9 Define impulse and show that how it is related to linear momentum?

Impulse is the product of a force and the time for which it acts: Impulse = F x t. By Newton's second law, F = m a = m (vf - vi) / t, so F x t = m vf - m vi, which is the change in linear momentum. Impulse is therefore equal to the change in linear momentum, and its unit is N s, the same as kg ms^-1.

3.10 State the law of conservation of linear momentum, pointing out the importance

of isolated system. Explain, why under certain conditions, the law is useful even though the system is not completely isolated?

The law of conservation of linear momentum states that the total linear momentum of an isolated system remains constant, provided no external force acts on the system. An isolated system is important because only then is the total external impulse zero, so the total momentum cannot change. The law is still useful for systems that are not completely isolated when the external forces are negligible during a very short interaction, such as a collision or an explosion, where the large internal forces act for such a short time that the external impulse is negligibly small, or when the external forces cancel out in the direction being considered.

3.11 Explain the difference between elastic and inelastic collisions. Explain how would a bouncing ball behave in each case? Give plausible reasons for the fact that K.E is not conserved in most cases?

In an elastic collision both momentum and kinetic energy are conserved, and the colliding bodies separate without permanent change. In an inelastic collision momentum is conserved but kinetic energy is not conserved; part of it is changed into heat, sound and energy of deformation. A bouncing ball in a nearly elastic bounce returns almost to its original height each time, while in an inelastic bounce it returns to a lower height each time and soon stops bouncing. Kinetic energy is not conserved in most real collisions because real bodies deform, vibrate and warm up on impact, so kinetic energy is converted into internal energy, sound and heat.

3.12 Explain what is meant by projectile motion. Derive expressions for (a) the time of flight (b) the range of projectile. Show that the range of projectile is maximum when projectile is thrown at an angle of 45 degrees with the horizontal.

Projectile motion is the motion of a body thrown into the air which then moves under the action of gravity alone, following a parabolic path. (a) Time of flight: the vertical motion gives T = 2 vi sin(theta) / g, where vi is the initial speed and theta

the angle of projection. (b) Range of projectile: the horizontal distance covered in this time is R = vi cos(theta) x T = vi^2 sin(2 theta) / g. The range is maximum when sin(2 theta) is maximum, that is, when sin(2 theta) = 1, which gives 2 theta = 90 degrees, so theta = 45 degrees.

3.13 At what point or points in its path does a projectile have its minimum speed, its maximum speed?

The speed of a projectile is minimum at the highest point of its path, because there its vertical component of velocity is zero and only the horizontal component, vi cos(theta), remains. The speed is maximum at the points of projection and of landing (at the same level), where the speed is the full initial speed vi, because the vertical component is largest there in size.

3.14 Each of the following questions is followed by four answers, one of which is correct answer. Identify that answer. (i) What is meant by a ballistic trajectory? (a) The paths followed by an un-powered and unguided projectile. (b) The path followed by the powered and unguided projectile. (c) The path followed by un-powered but guided projectile. (d) The path followed by powered and guided projectile. (ii) What happens when a system of two bodies undergoes an elastic collision? (a) The momentum of the system changes. (b) The momentum of the system does not change. (c) The bodies come to rest after collision. (d) The energy conservation law is violated.

(i) The correct answer is (a): a ballistic trajectory is the path followed by an un-powered and unguided projectile, which moves under gravity alone after launch.

(ii) The correct answer is (b): when a system of two bodies undergoes an elastic collision, the momentum of the system does not change; in fact both momentum and kinetic energy are conserved, so the bodies do not have to come to rest and no conservation law is violated.

Exercise 3 - Numerical Problems

3.1 A helicopter is ascending vertically at the rate of 19.6 ms^-1. When it is at a height of 156.8 m above the ground, a stone is dropped. How long does the stone take to reach the ground?

Step 1 - The stone starts with the helicopter's upward velocity, 19.6 ms^-1 upward. Take downward as positive: initial velocity u = -19.6 ms^-1, acceleration a = +9.8 ms^-2, distance s = 156.8 m.

Step 2 - Use s = u t + (1/2) a t^2: 156.8 = -19.6 t + 4.9 t^2.

Step 3 - Divide by 4.9: 32 = -4 t + t^2, so t^2 - 4 t - 32 = 0.

Step 4 - Factorise: (t - 8)(t + 4) = 0. The positive root is t = 8 s.

Answer: 8.0 s (as printed in the book).

3.2 Using the following data, draw a velocity-time graph for a short journey on a straight road of a motorbike. Velocity (ms^-1): 0, 10, 20, 20, 20, 20, 0 at Time (s): 0, 30, 60, 90, 120, 150, 180. Use the graph to calculate (a) the initial acceleration (b) the final acceleration and (c) the total distance travelled by the motorcyclist.

Step 1 - Graph instruction: plot velocity on the vertical axis against time on the horizontal axis, join the points (0,0), (30,10), (60,20), (90,20), (120,20), (150,20), (180,0) with straight lines. The graph rises steadily to 20 ms^-1, stays level, then falls to zero.

Step 2 - (a) Initial acceleration = slope of the rising part = (20 - 0) / (60 - 0) = 20 / 60 = 0.33 ms^-2.

Step 3 - (b) Final acceleration = slope of the falling part = (0 - 20) / (180 - 150) = -20 / 30 = -0.67 ms^-2.

Step 4 - (c) Total distance = area under the velocity-time graph. From 0 to 30 s: (1/2) x 30 x 10 = 150 m. From 30 to 60 s: (1/2) x (10 + 20) x 30 = 450 m.

Step 5 - From 60 to 150 s: 20 x 90 = 1800 m. From 150 to 180 s: (1/2) x 20 x 30 = 300 m.

Step 6 - Total = 150 + 450 + 1800 + 300 = 2700 m = 2.7 km.

Answer: (a) 0.33 ms^-2, (b) -0.67 ms^-2, (c) 2.7 km (as printed in the book).

3.3 A proton moving with speed of 1.0 x 10^7 ms^-1 passes through a 0.020 cm thick sheet of paper and emerges with a speed of 2.0 x 10^6 ms^-1. Assuming uniform deceleration, find retardation and time taken to pass through the paper.

Step 1 - Thickness in metres: s = 0.020 cm = 2.0 x 10^-4 m.

Step 2 - Use vf^2 - vi^2 = 2 a s: (2.0 x 10^6)^2 - (1.0 x 10^7)^2 = 2 a x (2.0 x 10^-4).

Step 3 - Left side: 4.0 x 10^12 - 1.0 x 10^14 = -9.6 x 10^13.

Step 4 - So a = (-9.6 x 10^13) / (4.0 x 10^-4) = -2.4 x 10^17 ms^-2. The retardation is 2.4 x 10^17 ms^-2.

Step 5 - Time: t = (vf - vi) / a = (2.0 x 10^6 - 1.0 x 10^7) / (-2.4 x 10^17) = (-8.0 x

10^6) / (-2.4 x 10^17) = 3.3 x 10^-11 s.

Answer: 2.4 x 10^17 ms^-2, 3.3 x 10^-11 s (as printed in the book).

3.4 Two masses m1 and m2 are initially at rest with a spring compressed between them. What is the ratio of the magnitude of their velocities after the spring has been released?

Step 1 - Initially the total momentum is zero, and no external force acts while the spring pushes the masses apart, so total momentum stays zero.

Step 2 - After release: m1 v1 + m2 v2 = 0 (with opposite signs for opposite directions).

Step 3 - In magnitudes: m1 v1 = m2 v2, so v1 / v2 = m2 / m1.

Answer: v1 / v2 = m2 / m1 (as printed in the book).

3.5 An amoeba of mass 1.0 x 10^-12 kg propels itself through water by blowing a jet of water through a tiny orifice. The amoeba ejects water with a speed of 1.0 x 10^-4 ms^-1 and at a rate of 1.0 x 10^-13 kgs^-1. Assume that the water is being continuously replenished so that the mass of the amoeba remains the same. (a) If there were no force on amoeba other than the reaction force caused by the emerging jet, what would be the acceleration of the amoeba? (b) If amoeba moves with constant velocity through water, what is force of surrounding water (exclusively of jet) on the amoeba?

Step 1 - Thrust of the jet = rate of mass ejection x ejection speed = (1.0 x 10^-13 kgs^-1) x (1.0 x 10^-4 ms^-1) = 1.0 x 10^-17 N.

Step 2 - (a) Acceleration = force / mass = (1.0 x 10^-17) / (1.0 x 10^-12) = 1.0 x 10^-5 ms^-2.

Step 3 - (b) At constant velocity the net force is zero, so the resistance of the surrounding water must exactly balance the thrust: 1.0 x 10^-17 N, opposite to the thrust.

Answer: (a) 1.0 x 10^-5 ms^-2 (b) 1.0 x 10^-17 N (as printed in the book).

3.6 A boy places a fire cracker of negligible mass in an empty can of 40 g mass. He plugs the end with a wooden block of mass 200 g. After igniting the firecracker, he throws the can straight up. It explodes at the top of its path. If the block shoots out with a speed of 3.0 ms^-1, how fast will the can be going?

Step 1 - At the top of its path the can and block are momentarily at rest, so the total momentum just before the explosion is zero.

Step 2 - After the explosion, momentum is conserved: (0.200 kg x 3.0 ms^-1) + (0.040 kg x v) = 0.

Step 3 - So 0.040 v = -0.600, giving v = -15 ms^-1. The minus sign means the can moves in the opposite direction to the block.

Answer: 15 ms^-1 (as printed in the book).

3.7 An electron (m = 9.1 x 10^-31 kg) travelling at 2.0 x 10^7 ms^-1 undergoes a head on collision with a hydrogen atom (m = 1.67 x 10^-27 kg) which is initially at rest. Assuming the collision to be perfectly elastic and a motion to be along a

straight line, find the velocity of hydrogen atom.

Step 1 - For a perfectly elastic head-on collision with the target at rest, the velocity of the target is v2 = [2 m1 / (m1 + m2)] x u1.

Step 2 - v2 = [2 x 9.1 x 10^-31 / (9.1 x 10^-31 + 1.67 x 10^-27)] x 2.0 x 10^7.

Step 3 - The denominator is about 1.671 x 10^-27, so the bracket is (1.82 x 10^-30) / (1.671 x 10^-27) = 1.089 x 10^-3.

Step 4 - v2 = 1.089 x 10^-3 x 2.0 x 10^7 = 2.18 x 10^4 ms^-1, about 2.2 x 10^4 ms^-1.

Answer: 2.2 x 10^4 ms^-1 (as printed in the book).

3.8 A truck weighing 2500 kg and moving with a velocity of 21 ms^-1 collides with stationary car weighing 1000 kg. The truck and the car move together after the impact. Calculate their common velocity.

Step 1 - This is a completely inelastic collision, so momentum is conserved but the bodies move together afterwards.

Step 2 - Initial momentum = 2500 x 21 + 1000 x 0 = 52,500 kg ms^-1.

Step 3 - Total mass afterwards = 2500 + 1000 = 3500 kg.

Step 4 - Common velocity = 52,500 / 3500 = 15 ms^-1.

Answer: 15 ms^-1 (as printed in the book).

3.9 Two blocks of masses 2.0 kg and 0.50 kg are attached at the two ends of a compressed spring. The elastic potential energy stored in the spring is 10 J. Find the velocities of the blocks if the spring delivers its energy to the blocks when released.

Step 1 - Momentum is conserved and starts at zero: 2.0 v1 + 0.50 v2 = 0, so v2 = -4 v1.

Step 2 - Energy is conserved: (1/2) x 2.0 x v1^2 + (1/2) x 0.50 x v2^2 = 10 J.

Step 3 - Substitute v2 = -4 v1: v1^2 + 0.25 x 16 v1^2 = 10, so 5 v1^2 = 10 and v1^2 = 2.

Step 4 - v1 = 1.4 ms^-1 and v2 = -4 x 1.4 = -5.6 ms^-1. The opposite signs show the blocks move in opposite directions.

Answer: 1.4 ms^-1, -5.6 ms^-1 (as printed in the book).

3.10 A foot ball is thrown upward with an angle of 30 degrees with respect to the horizontal. To throw a 40 m pass what must be the initial speed of the ball?

Step 1 - Range R = vi^2 sin(2 theta) / g, with R = 40 m and theta = 30 degrees, so sin(2 theta) = sin(60 degrees) = 0.866.

Step 2 - vi^2 = R g / sin(2 theta) = (40 x 9.8) / 0.866 = 392 / 0.866 = 452.7.

Step 3 - vi = sqrt(452.7) = 21.3 ms^-1, about 21 ms^-1.

Answer: 21 ms^-1 (as printed in the book).

3.11 A ball is thrown horizontally from a height of 10 m with velocity of 21 ms^-1.

How far off it hit the ground and with what velocity?

Step 1 - Time to fall: vertical motion starts from rest, h = (1/2) g t^2, so t = sqrt(2 h / g) = sqrt(20 / 9.8) = sqrt(2.04) = 1.43 s.

Step 2 - Horizontal distance = horizontal speed x time = 21 x 1.43 = 30.0 m.

Step 3 - Vertical speed at the ground: vy = g t = 9.8 x 1.43 = 14.0 ms^-1.

Horizontal speed is still 21 ms^-1.

Step 4 - Resultant speed = sqrt(21^2 + 14.0^2) = sqrt(441 + 196) = sqrt(637) = 25.2 ms^-1, about 25 ms^-1.

Answer: 30 m, 25 ms^-1 (as printed in the book).

3.12 A bomber dropped a bomb at a height of 490 m when its velocity along the horizontal was 300 kmh^-1. (a) How long was it in air? (b) At what distance from the point vertically below the bomber at the instant the bomb was dropped, did it strike the ground?

Step 1 - Horizontal speed in ms^-1: 300 kmh^-1 = 300 x 1000 / 3600 = 83.33 ms^-1.

Step 2 - (a) Time in air from vertical fall: t = sqrt(2 h / g) = sqrt(980 / 9.8) = sqrt(100) = 10 s.

Step 3 - (b) Horizontal distance = 83.33 x 10 = 833 m.

Answer: 10 s, 833 m (as printed in the book).

3.13 Find the angle of projection of a projectile for which its maximum height and horizontal range are equal.

Step 1 - Maximum height H = vi^2 sin^2(theta) / (2 g). Horizontal range R = vi^2 sin(2 theta) / g = 2 vi^2 sin(theta) cos(theta) / g.

Step 2 - Set H = R: vi^2 sin^2(theta) / (2 g) = 2 vi^2 sin(theta) cos(theta) / g.

Step 3 - Cancel vi^2, sin(theta) and g: sin(theta) / 2 = 2 cos(theta), so sin(theta) = 4 cos(theta).

Step 4 - tan(theta) = 4, so theta = tan^-1(4) = 76 degrees.

Answer: 76 degrees (as printed in the book).

3.14 Prove that for angles of projection, which exceed or fall short of 45 degrees by equal amounts, the ranges are equal.

Step 1 - Range at angle theta: R = vi^2 sin(2 theta) / g.

Step 2 - For the angle 45 degrees + alpha: sin(2 theta) = sin(90 degrees + 2 alpha) = cos(2 alpha).

Step 3 - For the angle 45 degrees - alpha: sin(2 theta) = sin(90 degrees - 2 alpha) = cos(2 alpha).

Step 4 - Both angles give the same value of sin(2 theta), so both give the same

range, R = vi^2 cos(2 alpha) / g.

Answer: the ranges are equal.

3.15 A SLBM (submarine launched ballistic missile) is fired from a distance of 3000 km. If the Earth is considered flat and the angle of launch is 45 degrees with horizontal, find the velocity with which the missile is fired and the time taken by SLBM to hit the target.

Step 1 - Range R = 3000 km = 3.0 x 10^6 m, angle theta = 45 degrees, so sin(2 theta) = sin(90 degrees) = 1.

Step 2 - R = vi^2 / g, so vi^2 = R g = 3.0 x 10^6 x 9.8 = 2.94 x 10^7.

Step 3 - vi = sqrt(2.94 x 10^7) = 5422 ms^-1 = 5.42 kms^-1.

Step 4 - Time of flight T = 2 vi sin(theta) / g = 2 x 5422 x 0.7071 / 9.8 = 7668 / 9.8 = 782 s.

Step 5 - 782 s = 13.0 minutes, about 13 min.

Answer: 5.42 kms^-1, 13 min (as printed in the book).

Chapter 4 - Work and Energy

Exercise 4 - Questions

4.1 A person holds a bag of groceries while standing still, talking to a friend. A car is stationary with its engine running. From the stand point of work, how are these two situations similar?

In both cases no work is done in the physics sense. The person holds the bag still, so the upward force acts through zero displacement, and work = force x displacement = 0. The stationary car also has zero displacement, so the engine does no work on the car, even though it burns fuel and gives out heat. The similarity is that energy is being spent (by muscles, by fuel) but no mechanical work is done because nothing moves.

4.2 Calculate the work done in kilo joules in lifting a mass of 10 kg (at a steady velocity) through a vertical height of 10 m.

Work done against gravity = m g h = 10 kg x 9.8 ms^-2 x 10 m = 980 J. In kilo joules this is 0.98 kJ, about 1.0 kJ. At a steady velocity the lifting force equals the weight, so no extra work is done in accelerating the mass.

4.3 A force F acts through a distance L. The force is then increased to 3 F, and then acts through a further distance of 2 L. Draw the work diagram to scale.

Draw a force-distance graph with distance on the horizontal axis and force on the vertical axis. From distance 0 to L the force is F, giving a rectangle of area F x L. From distance L to 3L the force is 3F, giving a rectangle of area 3F x 2L = 6 F L. The total work is the total area under the graph: F L + 6 F L = 7 F L. The second rectangle is three times as high and twice as wide as the first.

4.4 In which case is more work done? When a 50 kg bag of books is lifted through 50 cm, or when a 50 kg crate is pushed through 2 m across the floor with a force of 50 N?

More work is done in lifting the bag. Lifting: work = weight x height = 50 x 9.8 x 0.50 = 245 J. Pushing: work = force x distance = 50 N x 2 m = 100 J. Although the pushing distance is larger, the lifting force (the weight, 490 N) is much larger than

the pushing force, so the work done in lifting, 245 J, is greater than the 100 J done in pushing.

4.5 An object has 1 J of potential energy. Explain what does it mean?

It means the body has the capacity to do 1 joule of work by virtue of its position, for example by falling back to the chosen reference level. One joule of potential energy is stored energy that can be converted into kinetic energy or used to do work.

4.6 A ball of mass m is held at a height h1 above a table. The table top is at a height h2 above the floor. One student says that the ball has potential energy mgh1 but another says that it is mg (h1 + h2). Who is correct?

Both students can be correct, because potential energy is measured from an arbitrary reference level. Measured from the table top, the ball is at height h1, so its potential energy is m g h1. Measured from the floor, the ball is at height h1 + h2, so its potential energy is m g (h1 + h2). Only differences of potential energy have physical meaning; the zero level can be chosen freely.

4.7 When a rocket re-enters the atmosphere, its nose cone becomes very hot. Where does this heat energy come from?

The heat energy comes from the mechanical energy of the rocket itself. As the rocket rushes through the air, air friction (drag) does negative work on it, converting its kinetic energy, and some of its potential energy, into heat energy, which warms the nose cone and the air around it.

4.8 What sort of energy is in the following: (a) Compressed spring (b) Water in a high dam (c) A moving car

(a) Compressed spring: elastic potential energy, stored by the deformation of the spring. (b) Water in a high dam: gravitational potential energy, stored by virtue of its height above the turbines. (c) A moving car: kinetic energy, possessed by virtue of its motion.

4.9 A girl drops a cup from a certain height, which breaks into pieces. What energy

changes are involved?

First the cup has gravitational potential energy. As it falls, this changes into kinetic energy. On hitting the ground, the kinetic energy changes into sound energy, heat energy, and the energy needed to break the bonds of the material, that is, the energy of fracture and deformation of the pieces, with a little kinetic energy left in the flying pieces.

4.10 A boy uses a catapult to throw a stone which accidentally smashes a green house window. List the possible energy changes.

The boy's muscles do work to stretch the rubber of the catapult, storing elastic potential energy in it. On release, this changes into kinetic energy of the stone. As the stone flies, kinetic energy changes partly into gravitational potential energy. On striking the window, the kinetic energy changes into sound energy, heat energy, and the work done in breaking the glass, with kinetic energy carried away by the glass pieces.

Exercise 4 - Numerical Problems

4.1 A man pushes a lawn mower with a 40 N force directed at an angle of 20 degrees downward from the horizontal. Find the work done by the man as he cuts a strip of grass 20 m long.

Step 1 - Only the horizontal component of the force does work along the strip: F cos(theta) = 40 x cos(20 degrees).

Step 2 - cos(20 degrees) = 0.9397, so the horizontal component is 40 x 0.9397 = 37.59 N.

Step 3 - Work = component along displacement x distance = 37.59 x 20 = 751.8 J.

Answer: 7.5 x 10^2 J (as printed in the book).

4.2 A rain drop (m = 3.35 x 10^-5 kg) falls vertically at a constant speed under the influence of the forces of gravity and friction. In falling through 100 m, how much work is done by (a) gravity and (b) friction.

Step 1 - (a) Work by gravity = weight x distance fallen = m g h = 3.35 x 10^-5 x 9.8 x 100.

Step 2 - Work by gravity = 0.03283 J, about 0.0328 J. Gravity acts in the direction of motion, so this work is positive.

Step 3 - (b) At constant speed the net work is zero, so the work by friction must cancel the work by gravity.

Step 4 - Work by friction = -0.0328 J. Friction acts opposite to the motion, so its work is negative.

Answer: (a) 0.0328 J (b) -0.0328 J (as printed in the book).

4.3 Ten bricks, each 6.0 cm thick and mass 1.5 kg, lie flat on a table. How much work is required to stack them one on the top of another?

Step 1 - The first brick stays on the table and needs no lifting. The second brick is lifted by one thickness, the third by two thicknesses, and so on up to the tenth brick, lifted by nine thicknesses.

Step 2 - Total height lifted, summed over bricks = 6.0 cm x (1 + 2 + ... + 9) = 0.06 m x 45 = 2.70 m.

Step 3 - Work = m g x total height = 1.5 kg x 9.8 ms^-2 x 2.70 m = 39.7 J.

Answer: 40 J (as printed in the book).

4.4 A car of mass 800 kg travelling at 54 kmh^-1 is brought to rest in 60 metres. Find the average retarding force on the car. What has happened to original kinetic energy?

Step 1 - Speed in ms^-1: 54 kmh^-1 = 54 x 1000 / 3600 = 15 ms^-1.

Step 2 - By the work-energy principle, work done by the retarding force = loss of kinetic energy: F x s = (1/2) m v^2.

Step 3 - F x 60 = (1/2) x 800 x 15^2 = 90,000 J.

Step 4 - F = 90,000 / 60 = 1500 N.

Step 5 - The original kinetic energy is converted into heat energy in the brakes, tyres and road, with a little sound energy.

Answer: 1500 N (as printed in the book).

4.5 A 1000 kg automobile at the top of an incline 10 metre high and 100 m long is released and rolls down the hill. What is its speed at the bottom of the incline if the average retarding force due to friction is 480 N?

Step 1 - Potential energy at the top = m g h = 1000 x 9.8 x 10 = 98,000 J.

Step 2 - Work done against friction along the incline = force x distance = 480 x 100

= 48,000 J.

Step 3 - Kinetic energy at the bottom = 98,000 - 48,000 = 50,000 J.

Step 4 - (1/2) m v^2 = 50,000, so v^2 = 2 x 50,000 / 1000 = 100.

Step 5 - v = 10 ms^-1.

Answer: 10 ms^-1 (as printed in the book).

4.6 100 m^3 of water is pumped from a reservoir into a tank, 10 m higher than the reservoir, in 20 minutes. If density of water is 1000 kg m^-3, find (a) the increase in P.E. (b) the power delivered by the pump.

Step 1 - Mass of water = volume x density = 100 x 1000 = 100,000 kg.

Step 2 - (a) Increase in potential energy = m g h = 100,000 x 9.8 x 10 = 9.8 x 10^6 J.

Step 3 - (b) Time = 20 minutes = 1200 s. Power = energy / time = 9.8 x 10^6 / 1200 = 8167 W.

Step 4 - In kilowatts this is 8.2 kW.

Answer: (a) 9.8 x 10^6 J (b) 8.2 kW (as printed in the book).

4.7 A force (thrust) of 400 N is required to overcome road friction and air resistance in propelling an automobile at 80 kmh^-1. What power (kW) must the engine

develop?

Step 1 - Speed in ms^-1: 80 kmh^-1 = 80 x 1000 / 3600 = 22.22 ms^-1.

Step 2 - Power = force x velocity = 400 x 22.22 = 8889 W.

Step 3 - In kilowatts: 8.9 kW.

Answer: 8.9 kW (as printed in the book).

4.8 How large a force is required to accelerate an electron (m = 9.1 x 10^-31 kg) from rest to a speed of 2.0 x 10^7 ms^-1 through a distance of 5.0 cm?

Step 1 - By the work-energy principle: F x s = (1/2) m v^2.

Step 2 - (1/2) m v^2 = (1/2) x 9.1 x 10^-31 x (2.0 x 10^7)^2 = (1/2) x 9.1 x 10^-31 x 4.0 x 10^14 = 1.82 x 10^-16 J.

Step 3 - Distance s = 5.0 cm = 0.050 m, so F = 1.82 x 10^-16 / 0.050 = 3.64 x 10^-15 N.

Answer: 3.6 x 10^-15 N (as printed in the book).

4.9 A diver weighing 750 N dives from a board 10 m above the surface of a pool of water. Use the conservation of mechanical energy to find his speed at a point 5.0 m above the water surface, neglecting air friction.

Step 1 - Take the point 5.0 m above the water as the level to find the speed. The diver has fallen 10 - 5.0 = 5.0 m from the board.

Step 2 - Loss of potential energy = gain of kinetic energy: m g x 5.0 = (1/2) m v^2.

Step 3 - The mass cancels: v^2 = 2 x 9.8 x 5.0 = 98.

Step 4 - v = sqrt(98) = 9.9 ms^-1.

Answer: 9.9 ms^-1 (as printed in the book).

4.10 A child starts from rest at the top of a slide of height 4.0 m. (a) What is his speed at the bottom if the slide is frictionless? (b) If he reaches the bottom, with a speed of 6 ms^-1, what percentage of his total energy at the top of the slide is lost as a result of friction?

Step 1 - (a) Without friction, potential energy at the top becomes kinetic energy at the bottom: (1/2) m v^2 = m g h.

Step 2 - v^2 = 2 x 9.8 x 4.0 = 78.4, so v = sqrt(78.4) = 8.85 ms^-1, about 8.8 ms^-1.

Step 3 - (b) With friction the speed at the bottom is 6 ms^-1. Kinetic energy at the bottom is proportional to v^2, so the fraction remaining is (6 / 8.85)^2 = 36 / 78.4 = 0.459.

Step 4 - Fraction lost = 1 - 0.459 = 0.541, that is, 54 percent.

Answer: (a) 8.8 ms^-1 (b) 54% (as printed in the book).

Chapter 5 - Circular Motion

Exercise 5 - Questions

5.1 Explain the difference between tangential velocity and the angular velocity. If one of these is given for a wheel of known radius, how will you find the other?

Tangential velocity is the linear velocity of a point on a rotating body, directed along the tangent to its circular path; it is different for points at different distances from the axis. Angular velocity is the rate at which the body turns, the angle swept per unit time; it is the same for every point of the body. They are related by v = r x omega, where r is the distance from the axis. So if v is given for a wheel of known radius, omega = v / r; if omega is given, v = r x omega.

5.2 Explain what is meant by centripetal force and why it must be furnished to an object if the object is to follow a circular path?

Centripetal force is the net force directed towards the centre of a circular path, which is needed to keep a body moving in a circle. A body in circular motion is constantly changing the direction of its velocity, so it has an acceleration towards the centre, a = v^2 / r. By Newton's second law this acceleration needs a force, F = m v^2 / r, towards the centre. Without such a force the body would move off in a straight line along the tangent, by Newton's first law. The centripetal force can be supplied by tension in a string, gravity, friction, or the normal reaction, depending on the situation.

5.3 What is meant by moment of inertia? Explain its significance.

Moment of inertia is the rotational analogue of mass. It measures how much a body resists a change in its state of rotation, and it depends not only on the mass of the body but also on how that mass is distributed about the axis of rotation: I = sum of m r^2 over the body. Its significance is that, just as a larger mass needs a larger force for the same linear acceleration, a larger moment of inertia needs a larger

torque for the same angular acceleration, since torque = I x angular acceleration.

5.4 What is meant by angular momentum? Explain the law of conservation of angular momentum.

Angular momentum is the rotational analogue of linear momentum. For a body rotating about an axis it is L = I x omega, where I is the moment of inertia and omega the angular velocity; it is a vector along the axis of rotation. The law of conservation of angular momentum states that if no external torque acts on a system, its total angular momentum remains constant, because torque is the rate of change of angular momentum, so zero torque means zero change in angular momentum.

5.5 Show that orbital angular momentum L_o = m v r.

For a particle moving in a circle, the moment of inertia about the centre is I = m r^2, where m is its mass and r the radius of its path. Its angular velocity is omega = v / r, where v is its tangential speed. The orbital angular momentum is L = I x omega = m r^2 x (v / r) = m v r. This is also the magnitude of r cross p, since the radius and the momentum are perpendicular.

5.6 Describe what should be the minimum velocity, for a satellite, to orbit close to the Earth around it.

For a satellite in a low orbit close to the Earth, the gravitational force supplies the centripetal force: m v^2 / R = m g, where R is the radius of the Earth and g the acceleration due to gravity at its surface. So v = sqrt(g R) = sqrt(9.8 x 6.4 x 10^6) = 7.9 x 10^3 ms^-1, about 7.9 kms^-1. This is the minimum speed, the first cosmic velocity, for a satellite to stay in orbit just above the Earth.

5.7 State the direction of the following vectors in simple situations; angular momentum and angular velocity.

Angular velocity and angular momentum are vectors along the axis of rotation, in the direction given by the right-hand rule: curl the fingers of the right hand in the

sense of rotation, and the thumb points along the vector. For example, a wheel rotating anticlockwise when seen from above has its angular velocity and angular momentum pointing upward; rotating clockwise, they point downward. For the Earth spinning from west to east, they point towards the north pole; for a satellite orbiting eastward above the equator, they also point north.

5.8 Explain why an object, orbiting the Earth, is said to be freely falling. Use your explanation to point out why objects appear weightless under certain circumstances.

A satellite in orbit is continuously falling towards the Earth under gravity, but it also has a large tangential velocity, so as it falls it keeps missing the Earth and follows a curved path around it. This is free fall, like a projectile that never lands. Objects inside the satellite appear weightless because they too are falling with the same acceleration as the satellite, so there is no contact force between them and the floor of the satellite to register as weight. The same happens in a freely falling lift: everything inside falls together, so apparent weight becomes zero.

5.9 When mud flies off the tyre of a moving bicycle, in what direction does it fly? Explain.

The mud flies off along the tangent to the tyre at the point where it leaves the surface. While it is on the tyre, the tyre supplies the centripetal force that keeps it moving in a circle. Once it comes off, no such force acts on it, so by Newton's first law it continues in a straight line in the direction of its velocity at that instant, which is the tangent to the tyre.

5.10 A disc and a hoop start moving down from the top of an inclined plane at the same time. Which one will be moving faster on reaching the bottom?

The disc will be moving faster on reaching the bottom. Both start with the same potential energy, but part of it goes into rotational kinetic energy. The hoop has a larger moment of inertia for the same mass and radius (I = m r^2, against I = (1/2)

m r^2 for the disc), so more of its energy goes into rotation and less remains for translation. The acceleration down the plane is g sin(theta) / (1 + I / (m r^2)), which is (2/3) g sin(theta) for the disc but only (1/2) g sin(theta) for the hoop, so the disc accelerates more and arrives first.

5.11 Why does a diver change his body positions before and after diving in the pool?

A diver changes his body position to change his moment of inertia, and therefore his rate of rotation, while his angular momentum stays nearly constant in the air. Leaving the board he is stretched out, with a large moment of inertia and a slow rotation. In the air he tucks into a ball, reducing his moment of inertia, so his angular velocity increases and he somersaults quickly. Before entering the water he stretches out again, increasing his moment of inertia and slowing his rotation, so that he enters the water vertically.

5.12 A student holds two dumb-bells with stretched arms while sitting on a turn table. He is given a push until he is rotating at certain angular velocity. The student then pulls the dumb-bells towards his chest (Fig. 5.25). What will be the effect on rate of rotation?

The rate of rotation will increase. Pulling the dumb-bells towards his chest reduces the moment of inertia of the student about the vertical axis. Since no external torque acts, his angular momentum L = I x omega stays constant, so a smaller I must be balanced by a larger omega. The student therefore spins faster, as shown in Fig. 5.25, where the final angular velocity omega_f is greater than the initial omega_i.

5.13 Explain how many minimum number of geo-stationary satellites are required for global coverage of T.V transmission.

A single geo-stationary satellite, high above the equator, can see about one third of the Earth's surface, roughly 120 degrees of longitude. Three such satellites, spaced

120 degrees apart around the equator, are therefore the minimum number needed to cover the whole Earth (except the polar regions) for global television transmission, with each satellite handing over to the next as the Earth turns.

Exercise 5 - Numerical Problems

5.1 A tiny laser beam is directed from the Earth to the Moon. If the beam is to have a diameter of 2.50 m at the Moon, how small must divergence angle be for the beam? The distance of Moon from the Earth is 3.8 x 10^8 m.

Step 1 - For a small angle, angle in radians = arc length / radius = diameter at the Moon / distance to the Moon.

Step 2 - Angle = 2.50 / (3.8 x 10^8) = 6.58 x 10^-9 rad.

Answer: 6.6 x 10^-9 rad (as printed in the book).

5.2 A gramophone record turntable accelerates from rest to an angular velocity of 45.0 rev min^-1 in 1.60 s. What is its average angular acceleration?

Step 1 - Final angular velocity in rad s^-1: 45.0 rev min^-1 = 45.0 x 2 pi / 60 = 4.712 rad s^-1.

Step 2 - Average angular acceleration = change in angular velocity / time = (4.712 - 0) / 1.60 = 2.945 rad s^-2.

Answer: 2.95 rad s^-2 (as printed in the book).

5.3 A body of moment of inertia I = 0.80 kg m^2 about a fixed axis, rotates with a constant angular velocity of 100 rad s^-1. Calculate its angular momentum L and

the torque to sustain this motion.

Step 1 - Angular momentum L = I x omega = 0.80 x 100 = 80 J s.

Step 2 - The angular velocity is constant, so the angular acceleration is zero.

Step 3 - Torque = I x angular acceleration = 0.80 x 0 = 0. No torque is needed to sustain uniform rotation.

Answer: 80 J s, 0 (as printed in the book).

5.4 Consider the rotating cylinder shown in Fig. 5.26. Suppose that m = 5.0 kg, F = 0.60 N and r = 0.20 m. Calculate (a) the torque acting on the cylinder, (b) the angular acceleration of the cylinder. (Moment of inertia of cylinder = (1/2) m r^2.)

Step 1 - (a) Torque = force x radius = 0.60 x 0.20 = 0.12 N m.

Step 2 - Moment of inertia of the cylinder = (1/2) m r^2 = (1/2) x 5.0 x (0.20)^2 = 0.10 kg m^2.

Step 3 - (b) Angular acceleration = torque / moment of inertia = 0.12 / 0.10 = 1.2 rad s^-2.

Answer: 0.12 N m, 1.2 rad s^-2 (as printed in the book).

5.5 Calculate the angular momentum of a star of mass 2.0 x 10^30 kg and radius 7.0 x 10^5 km. If it makes one complete rotation about its axis once in 20 days, what is its kinetic energy?

Step 1 - Radius in metres: 7.0 x 10^5 km = 7.0 x 10^8 m. Period T = 20 days = 20

x 24 x 3600 = 1.728 x 10^6 s.

Step 2 - Angular velocity omega = 2 pi / T = 6.283 / (1.728 x 10^6) = 3.636 x 10^-6 rad s^-1.

Step 3 - Moment of inertia (sphere) I = (2/5) M R^2 = 0.4 x 2.0 x 10^30 x (7.0 x 10^8)^2 = 3.92 x 10^47 kg m^2.

Step 4 - Angular momentum L = I x omega = 3.92 x 10^47 x 3.636 x 10^-6 = 1.4 x 10^42 J s.

Step 5 - Rotational kinetic energy = (1/2) I omega^2 = (1/2) x 3.92 x 10^47 x (3.636 x 10^-6)^2 = 2.6 x 10^36 J, about 2.5 x 10^36 J.

Answer: 1.4 x 10^42 J s, 2.5 x 10^36 J (as printed in the book).

5.6 A 1000 kg car travelling with a speed of 144 km h^-1 round a curve of radius 100 m. Find the necessary centripetal force.

Step 1 - Speed in ms^-1: 144 km h^-1 = 144 x 1000 / 3600 = 40 ms^-1.

Step 2 - Centripetal force = m v^2 / r = 1000 x 40^2 / 100 = 1000 x 1600 / 100 = 16,000 N.

Answer: 1.60 x 10^4 N (as printed in the book).

5.7 What is the least speed at which an aeroplane can execute a vertical loop of 1.0 km radius so that there will be no tendency for the pilot to fall down at the highest

point?

Step 1 - At the highest point, the pilot's weight supplies the centripetal force when the seat just stops pushing on him: m g = m v^2 / r.

Step 2 - The mass cancels: v^2 = g r = 9.8 x 1000 = 9800.

Step 3 - v = sqrt(9800) = 99 ms^-1. Any slower, and the pilot would tend to fall away from his seat at the top.

Answer: 99 ms^-1 (as printed in the book).

5.8 The Moon orbits the Earth so that the same side always faces the Earth. Determine the ratio of its spin angular momentum (about its own axis) and its orbital angular momentum. (In this case, treat the Moon as a particle orbiting the Earth). Distance between the Earth and the Moon is 3.85 x 10^8 m. Radius of Moon is 1.74 x 10^6 m.

Step 1 - The Moon spins once per orbit, so its spin angular velocity equals its orbital angular velocity, omega.

Step 2 - Spin angular momentum = I x omega = (2/5) M R^2 x omega, with R = 1.74 x 10^6 m.

Step 3 - Orbital angular momentum (particle) = M x v x D = M x omega x D^2, with D = 3.85 x 10^8 m.

Step 4 - Ratio = [(2/5) M R^2 omega] / [M D^2 omega] = (2/5) x (R / D)^2 = 0.4 x (1.74 x 10^6 / 3.85 x 10^8)^2.

Step 5 - R / D = 4.519 x 10^-3. Its square is 2.042 x 10^-5. Ratio = 0.4 x 2.042 x 10^-5 = 8.2 x 10^-6.

Answer: 8.2 x 10^-6 (as printed in the book).

5.9 The Earth rotates on its axis once a day. Suppose, by some process the Earth contracts so that its radius is only half as large as at present. How fast will it be rotating then? (For sphere I = 2/5 M R^2.)

Step 1 - Moment of inertia of a sphere is proportional to R^2. Halving the radius reduces I to one quarter: I' = I / 4.

Step 2 - No external torque acts, so angular momentum is conserved: I' x omega' = I x omega.

Step 3 - omega' = omega x (I / I') = 4 omega. The Earth spins four times as fast.

Step 4 - New period = 24 hours / 4 = 6 hours.

Answer: The Earth would complete its rotation in 6 hours (as printed in the book).

5.10 What should be the orbiting speed to launch a satellite in a circular orbit 900 km above the surface of the Earth? (Take mass of the Earth as 6.0 x 10^24 kg and its radius as 6400 km).

Step 1 - Orbital radius from the Earth's centre: r = 6400 + 900 = 7300 km = 7.3 x 10^6 m.

Step 2 - Orbital speed v = sqrt(G M / r), with G = 6.67 x 10^-11 N m^2 kg^-2 and

M = 6.0 x 10^24 kg.

Step 3 - G M = 6.67 x 10^-11 x 6.0 x 10^24 = 4.00 x 10^14.

Step 4 - v = sqrt(4.00 x 10^14 / 7.3 x 10^6) = sqrt(5.48 x 10^7) = 7400 ms^-1 = 7.4 kms^-1.

Answer: 7.4 km s^-1 (as printed in the book).

Chapter 6 - Fluid Dynamics

Exercise 6 - Questions

6.1 Explain what do you understand by the term viscosity?

Viscosity is the internal friction of a fluid. It is the property by which one layer of a fluid resists motion relative to a neighbouring layer, so that a tangential force is needed to maintain flow. The coefficient of viscosity, eta, is defined as the tangential stress divided by the velocity gradient between layers. Its SI unit is N s m^-2, also called pascal second. Thick fluids like honey have a high viscosity, while water has a low viscosity.

6.2 What is meant by drag force? What are the factors upon which drag force acting upon a small sphere of radius r, moving down through a liquid, depend?

Drag force is the retarding force exerted by a fluid on a body moving through it, acting opposite to the direction of motion. For a small sphere of radius r moving slowly through a liquid, Stokes' law gives the drag force as F = 6 pi eta r v. The drag force therefore depends on the viscosity eta of the liquid, the radius r of the sphere,

and the speed v of the sphere. In general it also depends on the shape of the body and, at higher speeds, on the density of the fluid.

6.3 Why fog droplets appear to be suspended in air?

Fog droplets are extremely small, so their terminal velocity in air is extremely small. By Stokes' law the terminal velocity is proportional to the square of the radius, so a tiny droplet falls very slowly, only a fraction of a millimetre per second. The slightest upward currents of air, and even the random motion of air molecules, are enough to keep the droplets floating, so they appear to be suspended in the air.

6.4 Explain the difference between laminar flow and turbulent flow.

In laminar (streamline) flow, the fluid moves in smooth, orderly layers, each layer sliding past its neighbours without mixing, and the velocity at any point does not change with time. In turbulent flow, the motion is disorderly, with eddies and swirls, the velocity at a point keeps changing, and layers mix. Laminar flow occurs at low speeds; above a critical speed the flow becomes turbulent.

6.5 State Bernoulli's relation for a liquid in motion and describe some of its applications.

Bernoulli's relation for a liquid in steady, incompressible, non-viscous flow states that along a streamline, the sum of pressure energy, kinetic energy and potential energy per unit volume is constant: P + (1/2) rho v^2 + rho g h = constant, where P is pressure, rho density, v speed and h height. Its applications include the lift on an aeroplane wing, the Venturi meter for measuring flow speed, the carburettor of a motorcar, the filter pump and atomiser, the swing of a spinning cricket ball, and the design of chimneys and roofs in strong winds.

6.6 A person is standing near a fast moving train. Is there any danger that he will fall towards it?

Yes, there is a real danger. The fast train drags air with it, so the air between the person and the train moves fast, and by Bernoulli's principle its pressure falls. The

air behind the person is nearly still, at normal atmospheric pressure, so the pressure difference pushes the person towards the train. That is why passengers must stand well back from the edge of a platform.

6.7 Identify the correct answer. What do you infer from Bernoulli's theorem? (i) Where the speed of the fluid is high the pressure will be low. (ii) Where the speed of the fluid is high the pressure is also high. (iii) This theorem is valid only for turbulent flow of the liquid.

The correct answer is (i): where the speed of the fluid is high the pressure will be low. Statement (ii) is the opposite of Bernoulli's principle and is false. Statement (iii) is also false: Bernoulli's theorem applies to steady, streamline (laminar) flow of an incompressible, non-viscous fluid, not to turbulent flow, where energy is lost in eddies.

6.8 Two row boats moving parallel in the same direction are pulled towards each other. Explain.

The water between the two boats is confined to a narrow channel, and the boats drag it along, so it moves faster than the water on the outer sides. By Bernoulli's principle the pressure between the boats is therefore lower than the pressure outside them, and the higher outside pressure pushes the boats towards each other.

6.9 Explain, how the swing is produced in a fast moving cricket ball.

A cricket ball swings because the air flows differently on its two sides. One side of the ball is kept smooth and shiny, while the other becomes rough, and the seam is held at an angle. The boundary layer of air separates later on the smooth side, so the air speeds up more on one side than on the other. By Bernoulli's principle the pressure is lower on the faster side, and this sideways pressure difference pushes the ball sideways, making it swing in the air.

6.10 Explain the working of a carburetor of a motorcar using by Bernoulli's principle.

In a carburettor, air from outside is drawn through a narrow throat, a Venturi. As the

air passes the narrow part its speed increases and, by Bernoulli's principle, its pressure falls. A small jet opens into this low-pressure region, and the higher pressure on the petrol in the float chamber pushes petrol out of the jet as a fine spray. The spray mixes with the fast-moving air to form the combustible mixture that goes into the engine cylinders.

6.11 For which position will the maximum blood pressure in the body have the smallest value. (a) Standing up right (b) Sitting (c) Lying horizontally (d) Standing on one's head?

(c) Lying horizontally. When a person stands upright, the blood pressure in the feet is the heart pressure plus the hydrostatic pressure of the column of blood, rho g h, from the heart down to the feet, so the maximum pressure in the body is large. When the person lies horizontally, the heart, head and feet are all at nearly the same level, so there is almost no hydrostatic difference, and the maximum blood pressure in the body is at its smallest. Standing on one's head would make the pressure in the head the largest of all.

6.12 In an orbiting space station, would the blood pressure in major arteries in the leg ever be greater than the blood pressure in major arteries in the neck?

No. In an orbiting space station everything is in free fall, so there is no weight and no hydrostatic pressure difference due to gravity. On Earth the blood pressure in the legs is greater than in the neck because of the weight of the column of blood between them. In orbit that column has no weight, so the pressure in the major arteries of the legs will not be greater than in the neck; apart from the pumping of the heart, the pressures will be nearly equal.

Exercise 6 - Numerical Problems

6.1 Certain globular protein particle has a density of 1246 kg m^-3. It falls through pure water (eta = 8.0 x 10^-4 Nsm^-2) with a terminal speed of 3.0 cm h^-1. Find

the radius of the particle.

Step 1 - Terminal speed in ms^-1: 3.0 cm h^-1 = 0.030 m / 3600 s = 8.33 x 10^-6 ms^-1.

Step 2 - At terminal speed the viscous drag balances the weight minus the upthrust: 6 pi eta r v = (4/3) pi r^3 (rho - sigma) g, where sigma = 1000 kg m^-3 is the density of water.

Step 3 - This gives Stokes' terminal velocity: v = (2/9) x (rho - sigma) x g x r^2 / eta, so r^2 = 9 eta v / [2 (rho - sigma) g].

Step 4 - r^2 = 9 x 8.0 x 10^-4 x 8.33 x 10^-6 / [2 x (1246 - 1000) x 9.8] = 6.0 x 10^-8 / 4822 = 1.24 x 10^-11.

Step 5 - r = sqrt(1.24 x 10^-11) = 3.5 x 10^-6 m.

Note: the book prints the answer as 1.6 x 10^-6 m. That value follows if the upthrust of the water is left out, using the particle density alone: r^2 = 9 eta v / (2 rho g) = 6.0 x 10^-8 / 24422 = 2.46 x 10^-12, giving r = 1.6 x 10^-6 m. The physically complete Stokes result, with upthrust, is 3.5 x 10^-6 m; the book's printed answer is recorded here as printed.

6.2 Water flows through a hose, whose internal diameter is 1 cm at a speed of 1 ms^-1. What should be the diameter of the nozzle if the water is to emerge at 21 ms^-1?

Step 1 - By the equation of continuity, A1 v1 = A2 v2, and area is proportional to the square of the diameter.

Step 2 - So (d2 / d1)^2 = v1 / v2 = 1 / 21.

Step 3 - d2 = d1 x sqrt(1/21) = 1 cm x 0.218 = 0.218 cm, about 0.2 cm.

Answer: 0.2 cm (as printed in the book).

6.3 The pipe near the lower end of a large water storage tank develops a small leak and a stream of water shoots from it. The top of water in the tank is 15 m above the point of leak. (a) With what speed does the water rush from the hole? (b) If the hole has an area of 0.060 cm^2, how much water flows out in one second?

Step 1 - (a) By Torricelli's theorem, the speed of efflux is v = sqrt(2 g h) = sqrt(2 x 9.8 x 15) = sqrt(294) = 17.1 ms^-1.

Step 2 - (b) Volume flow rate = area x speed = 0.060 cm^2 x 1710 cm s^-1 (since 17.1 ms^-1 = 1710 cm s^-1).

Step 3 - Volume per second = 0.060 x 1710 = 102.6 cm^3, about 102 cm^3.

Answer: (a) 17 ms^-1, (b) 102 cm^3 (as printed in the book).

6.4 Water is flowing smoothly through a closed pipe system. At one point the speed of water is 3.0 ms^-1, while at another point 3.0 m higher, the speed is 4.0 ms^-1. If the pressure is 80 kPa at the lower point, what is pressure at the upper point?

Step 1 - Bernoulli's equation: P1 + (1/2) rho v1^2 + rho g h1 = P2 + (1/2) rho v2^2 + rho g h2. Take the lower point as h1 = 0, so h2 = 3.0 m.

Step 2 - P2 = P1 + (1/2) rho (v1^2 - v2^2) - rho g h2 = 80,000 + (1/2) x 1000 x (9 -

16) - 1000 x 9.8 x 3.0.

Step 3 - P2 = 80,000 - 3,500 - 29,400 = 47,100 Pa = 47.1 kPa.

Answer: 47 kPa (as printed in the book).

6.5 An airplane wing is designed so that when the speed of the air across the top of wing is 450 ms^-1, the speed of air below the wing is 410 ms^-1. What is the pressure difference between the top and bottom of the wings? (Density of air = 1.29 kgm^-3.)

Step 1 - Pressure difference = (1/2) rho (v_top^2 - v_bottom^2) = (1/2) x 1.29 x (450^2 - 410^2).

Step 2 - 450^2 - 410^2 = 202,500 - 168,100 = 34,400.

Step 3 - Pressure difference = 0.645 x 34,400 = 22,188 Pa, about 22 kPa. The pressure below the wing is higher, giving lift.

Answer: 22 kPa (as printed in the book).

6.6 The radius of the aorta is about 1.0 cm and the blood flowing through it has a speed of about 30 cms^-1. Calculate the average speed of the blood in the capillaries using the fact that although each capillary has a diameter of about 8 x 10^-4 cm, there are literally millions of them so that their total cross section is about 2000 cm^2.

Step 1 - Cross section of the aorta = pi r^2 = 3.14 x (1.0)^2 = 3.14 cm^2.

Step 2 - Volume flow rate = area x speed = 3.14 x 30 = 94.2 cm^3 s^-1.

Step 3 - By continuity, the same flow passes through the capillaries: speed in capillaries = flow rate / total capillary area = 94.2 / 2000 = 0.0471 cm s^-1.

Step 4 - In ms^-1 this is 4.71 x 10^-4 ms^-1, about 5 x 10^-4 ms^-1.

Answer: 5 x 10^-4 ms^-1 (as printed in the book).

6.7 How large must a heating duct be if air moving 3.0 ms^-1 along it can replenish the air in a room of 300 m^3 volume every 15 min? Assume the air's density remains constant.

Step 1 - Volume flow rate needed = 300 m^3 / (15 x 60 s) = 300 / 900 = 0.333 m^3 s^-1.

Step 2 - Area of duct = flow rate / speed = 0.333 / 3.0 = 0.111 m^2.

Step 3 - For a circular duct, area = pi r^2, so r = sqrt(0.111 / 3.14) = sqrt(0.0354) = 0.188 m, about 19 cm.

Answer: Radius = 19 cm (as printed in the book).

6.8 An airplane design calls for a 'lift' due to the net force of the moving air on the wing of about 1000 Nm^-2 of wing area. Assume that air flows past the wing of an aircraft with streamline flow. If the speed of flow past the lower wing surface is 160 ms^-1, what is the required speed over the upper surface to give a 'lift' of 1000 Nm^-2? The density of air is 1.29 kgm^-3 and assume maximum thickness of wing

to be one metre.

Step 1 - Lift per unit area = pressure difference = (1/2) rho (v_top^2 - v_bottom^2). The height term rho g x 1 m is only about 13 Pa and is negligible beside 1000 Pa, so it is left out.

Step 2 - 1000 = (1/2) x 1.29 x (v_top^2 - 160^2), so v_top^2 - 25,600 = 2000 / 1.29 = 1550.

Step 3 - v_top^2 = 25,600 + 1550 = 27,150, so v_top = sqrt(27,150) = 164.8 ms^-1, about 165 ms^-1.

Answer: 165 ms^-1 (as printed in the book).

6.9 What gauge pressure is required in the city mains for a stream from a fire hose connected to the mains to reach a vertical height of 15.0 m?

Step 1 - The gauge pressure must supply the potential energy of the water column: P = rho g h.

Step 2 - P = 1000 x 9.8 x 15.0 = 147,000 Pa = 1.47 x 10^5 Pa.

Answer: 1.47 x 10^5 Pa (as printed in the book).

Chapter 7 - Oscillations

Exercise 7 - Questions

7.1 Name two characteristics of simple harmonic motion.

Two characteristics of simple harmonic motion are: (1) the acceleration is directly proportional to the displacement from the mean position and is always directed towards the mean position, a = -omega^2 x; and (2) the motion is periodic, repeating itself after a fixed time, the time period, with the velocity greatest at the mean position and zero at the extreme positions.

7.2 Does frequency depends on amplitude for harmonic oscillators?

No. For a harmonic oscillator the frequency is fixed by the properties of the system, not by the amplitude. For a mass-spring system f = (1 / 2 pi) sqrt(k / m), and for a simple pendulum f = (1 / 2 pi) sqrt(g / l). As long as the motion remains simple harmonic, a larger amplitude only makes the body move faster, it does not change the time period. This independence of period from amplitude is called isochronism.

7.3 Can we realize an ideal simple pendulum?

An ideal simple pendulum is a point mass hanging from a light, inextensible string, swinging through small angles with no friction and no air resistance. It cannot be realised exactly in practice, because a real bob has a finite size, a real string has some mass and can stretch slightly, the support has friction, and air resistance always acts. A small, dense bob on a long, light thread swinging through a small angle is a good approximation to it.

7.4 What is the total distance travelled by an object moving with SHM in a time equal to its period, if its amplitude is A?

In one complete period the body travels four times the amplitude. Starting from the mean position it goes to one extreme (distance A), back to the mean position (distance A), on to the other extreme (distance A), and back to the mean position (distance A). The total distance is A + A + A + A = 4 A.

7.5 What happens to the period of a simple pendulum if its length is doubled? What happens if the suspended mass is doubled?

The period of a simple pendulum is T = 2 pi sqrt(l / g). If the length is doubled, the period becomes sqrt(2) times larger, about 1.41 times the original period. If the suspended mass is doubled, the period does not change at all, because the period of a simple pendulum does not depend on the mass.

7.6 Does the acceleration of a simple harmonic oscillator remain constant during its motion? Is the acceleration ever zero? Explain.

No, the acceleration does not remain constant, because a = -omega^2 x, so it changes with the displacement x: it is greatest at the extreme positions and changes sign on either side of the mean position. Yes, the acceleration is zero at the mean position, where x = 0, so the restoring force is zero there and the body momentarily has no acceleration, even though its velocity is greatest.

7.7 What is meant by phase angle? Does it define angle between maximum displacement and the driving force?

The phase angle is the quantity (omega t + phi) in the displacement equation y = A sin(omega t + phi). It tells the state of vibration of the body at any time, that is, how far through its cycle it has gone, and the initial phase phi is its value at t = 0. No, it does not by itself define the angle between the maximum displacement and the driving force; that phase difference belongs to forced vibrations, where the displacement lags behind the driving force by an amount that depends on the driving frequency and the damping.

7.8 Under what conditions does the addition of two simple harmonic motions produce a resultant, which is also simple harmonic?

The addition of two simple harmonic motions gives a resultant that is also simple harmonic when the two motions have the same frequency (the same period) and act along the same straight line. The resultant is then a simple harmonic motion of the same frequency, whose amplitude depends on the phase difference between the two motions: it is greatest when they are in phase and smallest when they are

exactly out of phase. If the frequencies differ, beats are produced, and if the motions are perpendicular, the path is a Lissajous figure, not a straight-line SHM.

7.9 Show that in SHM the acceleration is zero when the velocity is greatest and the velocity is zero when the acceleration is greatest?

In SHM the velocity is v = omega sqrt(A^2 - x^2) and the acceleration is a = -omega^2 x. At the mean position x = 0, so a = 0, while v = omega A, its greatest value. At an extreme position x = plus or minus A, so v = 0, while a = minus or plus omega^2 A, its greatest value in size. The acceleration is therefore zero exactly when the velocity is greatest, and the velocity is zero exactly when the acceleration is greatest.

7.10 In relation to SHM, explain the equations: (i) y = A sin (omega t + phi) (ii) a = -omega^2 x

(i) y = A sin(omega t + phi) gives the displacement y of the oscillator at time t. A is the amplitude, the maximum displacement; omega is the angular frequency, 2 pi times the frequency; and phi is the initial phase, the phase at t = 0, which fixes where in its cycle the body starts. (ii) a = -omega^2 x is the defining equation of SHM: the acceleration a is proportional to the displacement x, and the minus sign shows that it is directed opposite to the displacement, that is, always towards the mean position.

7.11 Explain the relation between total energy, potential energy and kinetic energy for a body oscillating with SHM.

The total energy of a body in SHM is constant: E = (1/2) m omega^2 A^2, where m is the mass, omega the angular frequency and A the amplitude. The potential energy is P.E. = (1/2) k x^2 = (1/2) m omega^2 x^2, which is greatest at the extremes and zero at the mean position. The kinetic energy is K.E. = (1/2) m v^2 = (1/2) m omega^2 (A^2 - x^2), which is greatest at the mean position and zero at the extremes. At every point, K.E. + P.E. = E, so energy continuously changes

between kinetic and potential forms while the total stays the same.

7.12 Describe some common phenomena in which resonance plays an important role.

Resonance is important in many common phenomena: a child on a swing goes higher when pushed at the swing's natural frequency; a radio or television set is tuned by making its circuit resonate with the frequency of the wanted station; musical instruments sound loudly because their strings or air columns resonate; a bridge can be damaged when soldiers march across it in step at its natural frequency, or when wind gusts match it, as in the collapse of the Tacoma Narrows bridge; buildings can suffer in an earthquake when the ground vibrations match their natural frequency; and a singer can shatter a glass by singing its resonant note.

7.13 If a mass spring system is hung vertically and set into oscillations, why does the motion eventually stop?

Because the motion is damped. Friction at the support, air resistance on the mass, and internal friction in the spring all oppose the motion and dissipate its energy as heat. The amplitude therefore decreases with every swing, and the oscillations gradually die away until the mass comes to rest at its equilibrium position.

Exercise 7 - Numerical Problems

7.1 A 100.0 g body hung on a spring elongates the spring by 4.0 cm. When a certain object is hung on the spring and set vibrating, its period is 0.568 s. What is the mass of the object pulling the spring?

Step 1 - Spring constant from the known body: k = m g / x = 0.100 x 9.8 / 0.040 = 24.5 N m^-1.

Step 2 - Period of a mass-spring system: T = 2 pi sqrt(m / k), so m = k x (T / 2 pi)^2.

Step 3 - m = 24.5 x (0.568 / 6.283)^2 = 24.5 x (0.0904)^2 = 24.5 x 0.00817 = 0.200 kg.

Answer: 0.20 kg (as printed in the book).

7.2 A load of 15.0 g elongates a spring by 2.00 cm. If body of mass 294 g is attached to the spring and is set into vibration with an amplitude of 10.0 cm, what will be its (i) period (ii) spring constant (iii) maximum speed of its vibration.

Step 1 - (ii) Spring constant: k = m g / x = 0.015 x 9.8 / 0.020 = 7.35 N m^-1.

Step 2 - (i) Period: T = 2 pi sqrt(m / k) = 2 pi sqrt(0.294 / 7.35) = 2 pi sqrt(0.040) = 2 pi x 0.20 = 1.26 s.

Step 3 - (iii) Angular frequency omega = sqrt(k / m) = sqrt(7.35 / 0.294) = sqrt(25) = 5.0 rad s^-1.

Step 4 - Maximum speed = omega x amplitude = 5.0 x 0.10 = 0.50 m s^-1 = 50 cm s^-1.

Answer: (i) 1.26 s, (ii) 7.35 N m^-1, (iii) 50 cm s^-1. Note: the book prints (iii) as 49.0 cm s^-1, a small difference from rounding the angular frequency from the rounded period; the direct value is 50 cm s^-1.

7.3 An 8.0 kg body executes SHM with amplitude 30 cm. The restoring force is 60 N when the displacement is 30 cm. Find (i) Period (ii) Acceleration, speed, kinetic energy and potential energy when the displacement is 12 cm.

Step 1 - Spring constant: k = F / x = 60 / 0.30 = 200 N m^-1.

Step 2 - (i) Period: T = 2 pi sqrt(m / k) = 2 pi sqrt(8 / 200) = 2 pi x 0.20 = 1.26 s, about 1.3 s.

Step 3 - Angular frequency omega = 2 pi / T = 5.0 rad s^-1.

Step 4 - (ii) Acceleration at x = 0.12 m: a = k x / m in size = 200 x 0.12 / 8 = 3.0 m s^-2, directed towards the mean position.

Step 5 - Speed at x = 0.12 m: v = omega sqrt(A^2 - x^2) = 5.0 x sqrt(0.30^2 - 0.12^2) = 5.0 x sqrt(0.0756) = 5.0 x 0.275 = 1.4 m s^-1.

Step 6 - Kinetic energy = (1/2) m v^2 = (1/2) x 8 x 1.4^2 = 7.6 J (using the unrounded speed 1.375 m s^-1 gives 7.56 J).

Step 7 - Potential energy = (1/2) k x^2 = (1/2) x 200 x (0.12)^2 = 100 x 0.0144 = 1.44 J.

Answer: (i) 1.3 s, (ii) 3.0 m s^-2, 1.4 m s^-1, 7.6 J, 1.44 J (as printed in the book).

7.4 A block of mass 4.0 kg is dropped from a height of 0.80 m on to a spring of spring constant k = 1960 Nm^-1. Find the maximum distance through which the spring will be compressed.

Step 1 - Let the maximum compression be x. The block falls a total distance (0.80 + x), so the loss of gravitational potential energy is m g (0.80 + x).

Step 2 - This equals the elastic potential energy stored at maximum compression: (1/2) k x^2. So 4.0 x 9.8 x (0.80 + x) = (1/2) x 1960 x x^2.

Step 3 - 39.2 (0.80 + x) = 980 x^2, that is, 980 x^2 - 39.2 x - 31.36 = 0.

Step 4 - Divide by 39.2: 25 x^2 - x - 0.80 = 0. Solving, x = [1 + sqrt(1 + 80)] / 50 = (1 + 9) / 50 = 0.20 m.

Note: the book prints 0.18 m. That value follows if the small extra fall x during compression is left out, using m g x 0.80 = (1/2) k x^2, which gives x = sqrt(2 x 39.2 x 0.80 / 1960) = 0.18 m. The complete energy result is 0.20 m; the book's printed answer is recorded here as printed.

7.5 A simple pendulum is 50.0 cm long. What will be its frequency of vibration at a place where g = 9.8 ms^-2?

Step 1 - Period T = 2 pi sqrt(l / g) = 2 pi sqrt(0.50 / 9.8) = 2 pi x 0.2259 = 1.42 s.

Step 2 - Frequency f = 1 / T = 1 / 1.42 = 0.705 Hz, about 0.70 Hz.

Answer: 0.70 Hz (as printed in the book).

7.6 A block of mass 1.6 kg is attached to a spring with spring constant 1000 Nm^-1, as shown in Fig. 7.14. The spring is compressed through a distance of 2.0 cm and the block is released from rest. Calculate the velocity of the block as it passes through the equilibrium position, x = 0, if the surface is frictionless.

Step 1 - Elastic potential energy at release = (1/2) k x^2 = (1/2) x 1000 x (0.020)^2 = 0.20 J.

Step 2 - At the equilibrium position all of this has become kinetic energy: (1/2) m v^2 = 0.20 J.

Step 3 - v^2 = 2 x 0.20 / 1.6 = 0.25, so v = 0.50 m s^-1.

Answer: 0.50 ms^-1 (as printed in the book).

7.7 A car of mass 1300 kg is constructed using a frame supported by four springs. Each spring has a spring constant 20,000 Nm^-1. If two people riding in the car have a combined mass of 160 kg, find the frequency of vibration of the car, when it is driven over a pot hole in the road. Assume the weight is evenly distributed.

Step 1 - Total mass = 1300 + 160 = 1460 kg. The four springs act in parallel, so the total spring constant is 4 x 20,000 = 80,000 N m^-1.

Step 2 - Frequency f = (1 / 2 pi) sqrt(k / m) = (1 / 6.283) x sqrt(80,000 / 1460).

Step 3 - sqrt(80,000 / 1460) = sqrt(54.79) = 7.40. So f = 7.40 / 6.283 = 1.18 Hz.

Answer: 1.18 Hz (as printed in the book).

7.8 Find the amplitude, frequency and period of an object vibrating at the end of a spring, if the equation for its position, as a function of time, is x = 0.25 cos ((pi / 8) t). What is the displacement of the object after 2.0 s?

Step 1 - Compare with x = A cos(omega t): amplitude A = 0.25 m and angular frequency omega = pi / 8 rad s^-1.

Step 2 - Frequency f = omega / (2 pi) = (pi / 8) / (2 pi) = 1 / 16 Hz.

Step 3 - Period T = 1 / f = 16 s.

Step 4 - Displacement after 2.0 s: x = 0.25 x cos((pi / 8) x 2.0) = 0.25 x cos(pi / 4) = 0.25 x 0.7071 = 0.177 m, about 0.18 m.

Answer: 0.25 m, (1/16) Hz, 16 s, x = 0.18 m (as printed in the book).

Chapter 8 - Waves

Exercise 8 - Questions

8.1 What features do longitudinal waves have in common with transverse waves?

Both longitudinal and transverse waves carry energy from one place to another without any net transport of the medium itself. Both can be described by the same quantities: amplitude, wavelength, frequency and period, with wave speed v = f lambda. Both obey the principle of superposition, and both show reflection, refraction, diffraction and interference. The difference is only in the direction of vibration of the particles relative to the direction of propagation.

8.2 The five possible waveforms obtained, when the output from a microphone is fed into the Y-input of cathode ray oscilloscope, with the time base on, are shown in Fig. 8.23. These waveforms are obtained under the same adjustment of the cathode ray oscilloscope controls. Indicate the waveform (a) which trace represents the loudest note? (b) which trace represents the highest frequency?

The loudness of a note depends on the amplitude of the waveform, and the frequency on the number of cycles in a given time, since the time base is the same for all traces. (a) The loudest note is trace D, which has the greatest amplitude, the tallest peaks. (b) The highest frequency is trace B, which shows the most complete cycles in the same time, so it has the most vibrations per second.

8.3 Is it possible for two identical waves travelling in the same direction along a string to give rise to a stationary wave?

No. A stationary wave is formed by the superposition of two identical waves travelling in opposite directions along the same string, so that the incident wave and its reflection combine to give fixed nodes and antinodes. Two identical waves travelling in the same direction simply add together to give a larger travelling wave, not a stationary wave.

8.4 A wave is produced along a stretched string but some of its particles permanently show zero displacement. What type of wave is it?

It is a stationary (standing) wave. In a stationary wave, some points, the nodes, are permanently at rest and always show zero displacement, while points between them vibrate with maximum amplitude at the antinodes.

8.5 Explain the terms crest, trough, node and antinode.

A crest is the highest point of a transverse wave, where the displacement is maximum in one direction. A trough is the lowest point, where the displacement is maximum in the opposite direction. A node is a point in a stationary wave where the particles are permanently at rest, with zero amplitude. An antinode is a point in a stationary wave midway between two nodes, where the particles vibrate with maximum amplitude.

8.6 Why does sound travel faster in solids than in gases?

The speed of sound in a medium depends on its elasticity and its density: v = sqrt(E / rho), where E is the elastic modulus. Solids are far more elastic than gases, and their particles are closely packed and strongly linked, so a disturbance is passed on very quickly from particle to particle. The much greater elasticity of solids more than makes up for their greater density, so sound travels faster in solids than in gases.

8.7 How are beats useful in tuning musical instruments?

When two notes of nearly the same frequency are sounded together, beats are heard, and the beat frequency equals the difference of the two frequencies. To tune an instrument, its note is sounded together with a standard note, such as a tuning fork of known frequency. The tension or length of the string is adjusted until the beats slow down and finally disappear, which shows that the two frequencies are now equal and the instrument is in tune.

8.8 When two notes of frequencies f1 and f2 are sounded together, beats are formed. If f1 > f2, what will be the frequency of beats? (i) f1 + f2 (ii) (1/2)(f1 + f2) (iii) f1 - f2 (iv) (1/2)(f1 - f2)

(iii) f1 - f2. The beat frequency is the difference between the two frequencies. Since f1 is greater than f2, the number of beats per second is f1 - f2.

8.9 As a result of a distant explosion, an observer senses a ground tremor and then hears the explosion. Explain the time difference.

The ground tremor is carried by seismic waves travelling through the Earth, which move much faster than sound waves travelling through the air. The observer therefore feels the tremor from the explosion before the sound of the explosion reaches his ears, and the time difference is due to the different speeds of the two kinds of wave over the same distance.

8.10 Explain why sound travels faster in warm air than in cold air.

The speed of sound in a gas increases with temperature, v being proportional to the square root of the absolute temperature. In warm air the molecules move faster and collide more often, so a disturbance is handed on more quickly from molecule to molecule. The speed of sound in air is given by v = v0 + 0.61 t, increasing by about 0.61 m s^-1 for every degree Celsius rise in temperature.

8.11 How should a sound source move with respect to an observer so that the frequency of its sound does not change?

The change in frequency, the Doppler effect, depends on the component of the

relative motion along the line joining the source and the observer. If the source moves at right angles to that line, or moves with exactly the same velocity as the observer so that there is no relative motion towards or away from the observer, the apparent frequency does not change.

Exercise 8 - Numerical Problems

8.1 The wavelength of the signals from a radio transmitter is 1500 m and the frequency is 200 kHz. What is the wavelength for a transmitter operating at 1000 kHz and with what speed the radio waves travel?

Step 1 - Speed of the radio waves: v = f lambda = 200 x 10^3 x 1500 = 3.0 x 10^8 m s^-1.

Step 2 - Radio waves all travel at the same speed, so for the second transmitter, wavelength = v / f = 3.0 x 10^8 / (1000 x 10^3) = 300 m.

Answer: 300 m, 3 x 10^8 ms^-1 (as printed in the book).

8.2 Two speakers are arranged as shown in Fig. 8.24. The distance between them is 3 m and they emit a constant tone of 344 Hz. A microphone P is moved along a line parallel to and 4.00 m from the line connecting the two speakers. It is found that tone of maximum loudness is heard and displayed on the CRO when microphone is on the centre of the line and directly opposite each speakers. Calculate the speed of sound.

Step 1 - At the centre of the line the microphone is equidistant from the two speakers, so the path difference is zero, the first maximum.

Step 2 - Directly opposite one speaker, the distance to that speaker is 4.00 m, and

to the other speaker it is sqrt(4.00^2 + 3^2) = sqrt(25) = 5.00 m.

Step 3 - The path difference there is 5.00 - 4.00 = 1.00 m. Since this is also a maximum, the path difference equals one wavelength: lambda = 1.00 m.

Step 4 - Speed of sound v = f lambda = 344 x 1.00 = 344 m s^-1.

Answer: 344 ms^-1 (as printed in the book).

8.3 A stationary wave is established in a string which is 120 cm long and fixed at both ends. The string vibrates in four segments, at a frequency of 120 Hz. Determine its wavelength and the fundamental frequency?

Step 1 - Each segment is half a wavelength long. Four segments fill the string, so 120 cm = 2 wavelengths.

Step 2 - Wavelength lambda = 120 / 2 = 60 cm = 0.60 m.

Step 3 - Speed of the wave v = f lambda = 120 x 0.60 = 72 m s^-1.

Step 4 - Fundamental frequency f1 = v / (2 L) = 72 / (2 x 1.20) = 72 / 2.40 = 30 Hz. The given vibration at 120 Hz is the fourth harmonic.

Answer: 0.6 m, 30 Hz (as printed in the book).

8.4 The frequency of the note emitted by a stretched string is 300 Hz. What will be the frequency of this note when; (a) the length of the wave is reduced by one-third without changing the tension. (b) the tension is increased by one-third without

changing the length of the wire.

Step 1 - (a) Frequency is inversely proportional to length: f is proportional to 1 / L. The new length is L - L/3 = (2/3) L.

Step 2 - New frequency = 300 x L / [(2/3) L] = 300 x 3 / 2 = 450 Hz.

Step 3 - (b) Frequency is proportional to the square root of tension: f is proportional to sqrt(T). The new tension is T + T/3 = (4/3) T.

Step 4 - New frequency = 300 x sqrt(4/3) = 300 x 1.155 = 346 Hz.

Answer: 450 Hz, 346 Hz (as printed in the book).

8.5 An organ pipe has a length of 50 cm. Find the frequency of its fundamental note and the next harmonic when it is (a) open at both ends. (b) closed at one end. (Speed of sound = 350 ms^-1.)

Step 1 - (a) Open at both ends: fundamental f1 = v / (2 L) = 350 / (2 x 0.50) = 350 Hz. The next harmonic is the second harmonic, 2 f1 = 700 Hz.

Step 2 - (b) Closed at one end: fundamental f1 = v / (4 L) = 350 / (4 x 0.50) = 175 Hz. The next harmonic is the third harmonic, 3 f1 = 525 Hz.

Answer: (a) 350 Hz, 700 Hz, (b) 175 Hz, 525 Hz (as printed in the book).

8.6 A church organ consists of pipes, each open at one end, of different lengths. The minimum length is 30 mm and the longest is 4 m. Calculate the frequency range of the fundamental notes. (Speed of sound = 340 ms^-1.)

Step 1 - Each pipe is closed at one end, so its fundamental is f1 = v / (4 L).

Step 2 - Longest pipe, L = 4 m: f1 = 340 / (4 x 4) = 340 / 16 = 21.25 Hz, about 21 Hz.

Step 3 - Shortest pipe, L = 30 mm = 0.030 m: f1 = 340 / (4 x 0.030) = 340 / 0.12 = 2833 Hz.

Answer: 21 Hz to 2833 Hz (as printed in the book).

8.7 Two tuning forks exhibit beats at a beat frequency of 3 Hz. The frequency of one fork is 256 Hz. Its frequency is then lowered slightly by adding a bit of wax to one of its prong. The two forks then exhibit a beat frequency of 1 Hz. Determine the frequency of the second tuning fork.

Step 1 - The beat frequency is the difference of the two frequencies, so the second fork is either 256 + 3 = 259 Hz or 256 - 3 = 253 Hz.

Step 2 - Wax is added to the 256 Hz fork, lowering its frequency slightly. If the second fork were 259 Hz, lowering the first fork would increase the difference, and the beat frequency would grow, not fall.

Step 3 - Since the beat frequency instead falls from 3 Hz to 1 Hz, the second fork must be below 256 Hz, so that lowering the first fork brings the two frequencies closer together.

Answer: 253 Hz (as printed in the book).

8.8 Two cars P and Q are travelling along a motorway in the same direction. The leading car P travels at a steady speed of 12 ms^-1; the other car Q, travelling at a steady speed of 20 ms^-1, sound its horn to emit a steady note which P's driver estimates, has a frequency of 830 Hz. What frequency does Q's own driver hear? (Speed of sound = 340 ms^-1.)

Step 1 - Q's driver travels with the source, so he hears the true frequency f of the horn. P is the moving observer.

Step 2 - Q is behind P and moving faster, so Q approaches P, while P moves away from Q. The Doppler formula gives f' = f x (v - v_observer) / (v - v_source) = f x (340 - 12) / (340 - 20).

Step 3 - 830 = f x 328 / 320, so f = 830 x 320 / 328 = 809.8 Hz, about 810 Hz.

Answer: 810 Hz (as printed in the book).

8.9 A train sounds its horn before it sets off from the station and an observer waiting on the platform estimates its frequency at 1200 Hz. The train then moves on and accelerates steadily. Fifty seconds after departure, the driver sounds the horn again and the platform observer estimates the frequency at 1140 Hz. Calculate the train speed 50 s after departure. How far from the station is the train after 50 s? (Speed of sound = 340 ms^-1.)

Step 1 - At rest the observer hears the true frequency, f = 1200 Hz.

Step 2 - For a source receding from a stationary observer, f' = f x v / (v + v_source). So 1140 = 1200 x 340 / (340 + v_source).

Step 3 - 340 + v_source = 1200 x 340 / 1140 = 357.9, so v_source = 17.9 m s^-1.

Step 4 - The train started from rest with steady acceleration, so its acceleration is a = v / t = 17.9 / 50 = 0.358 m s^-2.

Step 5 - Distance from the station: s = (1/2) a t^2 = (1/2) x 0.358 x 50^2 = 0.179 x 2500 = 447.5 m, about 448 m.

Answer: 17.9 ms^-1, 448 m (as printed in the book).

8.10 The absorption spectrum of faint galaxy is measured and the wavelength of one of the lines identified as the Calcium alpha line is found to be 478 nm. The same line has a wavelength of 397 nm when measured in a laboratory. (a) Is the galaxy moving towards or away from the Earth? (b) Calculate the speed of the galaxy relative to Earth. (Speed of light = 3.0 x 10^8 ms^-1.)

Step 1 - (a) The observed wavelength, 478 nm, is longer than the laboratory wavelength, 397 nm. The light is shifted towards the red end of the spectrum, a red shift, which means the galaxy is moving away from the Earth.

Step 2 - (b) For speeds much less than the speed of light, the fractional shift equals v / c: (delta lambda) / lambda = v / c.

Step 3 - Change in wavelength = 478 - 397 = 81 nm. So v = c x 81 / 397 = 3.0 x 10^8 x 0.204 = 6.12 x 10^7 m s^-1.

Answer: (a) away from the Earth, (b) 6.1 x 10^7 ms^-1 (as printed in the book).

Chapter 9 - Physical Optics

Exercise 9 - Questions

9.1 Under what conditions two or more sources of light behave as coherent sources?

Two or more sources of light behave as coherent sources when they emit waves of the same frequency and wavelength with a constant phase difference between them, so that the interference pattern does not shift with time. In practice this is achieved by dividing the light from a single source, for example by Young's double slit or by a biprism, because two independent sources change their relative phase randomly and cannot give a steady pattern.

9.2 How is the distance between interference fringes affected by the separation between the slits of Young's experiment? Can fringes disappear?

The distance between adjacent fringes is delta y = L lambda / d, where L is the slit-to-screen distance, lambda the wavelength and d the slit separation. The fringe spacing is therefore inversely proportional to the slit separation: increasing d makes the fringes closer together. Yes, the fringes can disappear: when d becomes so large that the fringes are too fine to be resolved by the eye, or so large that the two slits no longer act as coherent sources, the pattern washes out into uniform illumination.

9.3 Can visible light produce interference fringes? Explain.

Yes. Visible light produces interference fringes whenever coherent beams overlap, as in Young's double-slit experiment, where a screen shows alternate bright and dark fringes. Thin films also show it: soap bubbles and oil films on water display coloured fringes, and Newton's rings are interference fringes formed in the thin air film between a lens and a glass plate. The fringes are visible because the wavelength of light is small and the slit separation or film thickness is comparable with it.

9.4 In the Young's experiment, one of the slits is covered with blue filter and other with red filter. What would be the pattern of light intensity on the screen?

No interference pattern would appear. Light passing through the blue filter has a different wavelength from light passing through the red filter, so the two beams are not coherent and cannot interfere. The screen would show only the sum of the two separate single-slit diffraction patterns, a uniform overlapping wash of blue and red light with no bright and dark interference fringes.

9.5 Explain whether the Young's experiment is an experiment for studying interference or diffraction effects of light.

Young's experiment is an experiment for studying interference. The pattern of bright and dark fringes on the screen is produced by the superposition of the two coherent waves coming from the two slits. Diffraction is also involved, because each narrow slit diffracts the light, spreading it out so that the two beams overlap on the screen, but the fringe pattern itself is an interference effect.

9.6 An oil film spreading over a wet footpath shows colours. Explain how does it happen?

The oil forms a very thin film on the water. White light is reflected partly from the top surface of the film and partly from the bottom surface, and the two reflected beams interfere. The path difference between them depends on the thickness of the film, which varies as the oil spreads. For each thickness, some wavelengths interfere constructively and appear bright, while others interfere destructively and are missing, so the film shows different colours at different places.

9.7 Could you obtain Newton's rings with transmitted light? If yes, would the pattern be different from that obtained with reflected light?

Yes, Newton's rings can also be obtained with transmitted light, because the light transmitted through the thin air film also interferes. The pattern is, however, different from the reflected-light pattern: it is complementary, with a bright centre

instead of a dark centre, and the rings are much fainter, because the two transmitted beams have very unequal intensities, so the contrast is poor.

9.8 In the white light spectrum obtained with a diffraction grating, the third order image of a wavelength coincides with the fourth order image of a second wavelength. Calculate the ratio of the two wavelengths.

For a diffraction grating, d sin(theta) = n lambda. The third order image of one wavelength and the fourth order image of the second wavelength appear at the same angle, so 3 lambda1 = 4 lambda2, since d and sin(theta) are the same for both. The ratio of the two wavelengths is therefore lambda1 : lambda2 = 4 : 3.

9.9 How would you manage to get more orders of spectra using a diffraction grating?

The maximum order obtainable is n = d / lambda, since sin(theta) cannot exceed 1. To get more orders, increase the grating spacing d, that is, use a coarser grating with fewer lines per centimetre, or use light of shorter wavelength. A finer grating spreads the orders wider apart and improves resolution, but it reduces the number of orders, so the two aims must be balanced.

9.10 Why the polaroid sunglasses are better than ordinary sunglasses?

Ordinary sunglasses only darken the view by absorbing a fraction of all the light. Polaroid sunglasses do more: light reflected from a road or water surface is largely polarized in the horizontal plane, and the polaroid lens is oriented to block that horizontally polarized glare, while letting through the vertically polarized part of the light. They therefore cut the dazzling glare much more effectively than ordinary dark glasses, without making the whole scene equally dark.

9.11 How would you distinguish between un-polarized and plane-polarized lights?

Pass the light through a polarizer, such as a polaroid sheet, and rotate the polarizer. For unpolarized light the transmitted intensity stays the same at every orientation, because it contains vibrations in all directions. For plane-polarized light the

transmitted intensity varies: it is maximum when the polarizer's axis is parallel to the direction of polarization, and falls to zero (minimum) when the axis is at right angles to it.

9.12 Fill in the blanks. (i) According to _____ principle, each point on a wavefront acts as a source of secondary _____. (ii) In Young's experiment, the distance between two adjacent bright fringes for violet light is _____ than that for green light. (iii) The distance between bright fringes in the interference pattern _____ as the wavelength of light used increases. (iv) A diffraction grating is used to make a diffraction pattern for yellow light and then for red light. The distances between the red spots will be _____ than that for yellow light. (v) The phenomenon of polarization of light reveals that light waves are _____ waves. (vi) A polaroid is a commercial _____. (vii) A polaroid glass _____ glare of light produced at a road surface.

(i) According to Huygen's principle, each point on a wavefront acts as a source of secondary wavelets. (ii) In Young's experiment, the distance between two adjacent bright fringes for violet light is less than that for green light. (iii) The distance between bright fringes in the interference pattern increases as the wavelength of light used increases. (iv) A diffraction grating is used to make a diffraction pattern for yellow light and then for red light. The distances between the red spots will be greater than that for yellow light. (v) The phenomenon of polarization of light reveals that light waves are transverse waves. (vi) A polaroid is a commercial polarizer. (vii) A polaroid glass reduces glare of light produced at a road surface.

Exercise 9 - Numerical Problems

9.1 Light of wavelength 546 nm is allowed to illuminate the slits of Young's experiment. The separation between the slits is 0.10 mm and the distance of the screen from the slits where interference effects are observed is 20 cm. At what

angle the first minimum will fall? What will be the linear distance on the screen between adjacent maxima?

Step 1 - First minimum: d sin(theta) = (m + 1/2) lambda with m = 0, so sin(theta) = lambda / (2 d) = 546 x 10^-9 / (2 x 0.10 x 10^-3) = 546 x 10^-9 / (2.0 x 10^-4) = 2.73 x 10^-3.

Step 2 - For such a small angle, theta in radians is about sin(theta) = 2.73 x 10^-3 rad = 2.73 x 10^-3 x (180 / pi) degrees = 0.156 degrees, about 0.16 degrees.

Step 3 - Linear distance between adjacent maxima (fringe spacing) = L lambda / d = 0.20 x 546 x 10^-9 / (0.10 x 10^-3) = 1.092 x 10^-3 m = 1.09 mm, about 1.1 mm.

Answer: 0.16 degrees, 1.1 mm (as printed in the book).

9.2 Calculate the wavelength of light, which illuminates two slits 0.5 mm apart and produces an interference pattern on a screen placed 200 cm away from the slits. The first bright fringe is observed at a distance of 2.40 mm from the central bright image.

Step 1 - For a bright fringe, y = m L lambda / d, with m = 1 for the first bright fringe.

Step 2 - So lambda = y d / L = (2.40 x 10^-3 x 0.5 x 10^-3) / 2.00.

Step 3 - lambda = 1.20 x 10^-6 / 2.00 = 6.0 x 10^-7 m = 600 nm.

Answer: 600 nm (as printed in the book).

9.3 In a double slit experiment the second order maximum occurs at theta = 0.25 degrees. The wavelength is 650 nm. Determine the slit separation.

Step 1 - For a bright fringe, d sin(theta) = m lambda, with m = 2.

Step 2 - d = m lambda / sin(theta) = 2 x 650 x 10^-9 / sin(0.25 degrees).

Step 3 - sin(0.25 degrees) = 0.004363, so d = 1300 x 10^-9 / 0.004363 = 2.98 x 10^-4 m = 0.30 mm.

Answer: 0.30 mm (as printed in the book).

9.4 A monochromatic light of lambda = 588 nm is allowed to fall on the half silvered glass plate G1, in the Michelson Interferometer. If mirror M1 is moved through 0.233 mm, how many fringes will be observed to shift?

Step 1 - When the mirror moves a distance L, the path difference changes by 2 L, and one fringe shifts for each wavelength of path change: number of fringes m = 2 L / lambda.

Step 2 - m = 2 x 0.233 x 10^-3 / (588 x 10^-9) = 466 x 10^-6 / (588 x 10^-9) = 792.5.

Step 3 - About 792 fringes shift.

Answer: 792 (as printed in the book; the unrounded value is 792.5).

9.5 A second order spectrum is formed at an angle of 38.0 degrees when light falls normally on a diffraction grating having 5400 lines per centimetre. Determine

wavelength of the light used.

Step 1 - Grating spacing d = 1 cm / 5400 = 1.852 x 10^-6 m.

Step 2 - For a grating, d sin(theta) = n lambda, with n = 2.

Step 3 - lambda = d sin(theta) / n = 1.852 x 10^-6 x sin(38.0 degrees) / 2 = 1.852 x 10^-6 x 0.6157 / 2 = 5.70 x 10^-7 m = 570 nm.

Answer: 570 nm (as printed in the book).

9.6 A light is incident normally on a grating which has 2500 lines per centimetre. Compute the wavelength of a spectral line for which the deviation in second order is 15.0 degrees.

Step 1 - Grating spacing d = 1 cm / 2500 = 4.0 x 10^-6 m.

Step 2 - d sin(theta) = n lambda, with n = 2, so lambda = d sin(theta) / 2 = 4.0 x 10^-6 x sin(15.0 degrees) / 2.

Step 3 - sin(15.0 degrees) = 0.2588, so lambda = 4.0 x 10^-6 x 0.2588 / 2 = 5.18 x 10^-7 m = 518 nm.

Answer: 518 nm (as printed in the book).

9.7 Sodium light (lambda = 589 nm) is incident normally on a grating having 3000 lines per centimetre. What is the highest order of the spectrum obtained with this grating?

Step 1 - Grating spacing d = 1 cm / 3000 = 3.333 x 10^-6 m.

Step 2 - The highest order is reached when sin(theta) = 1: n_max = d / lambda = 3.333 x 10^-6 / (589 x 10^-9) = 5.66.

Step 3 - Only whole orders occur, so the highest order actually obtained is the 5th order.

Answer: 5th (as printed in the book).

9.8 Blue light of wavelength 480 nm illuminates a diffraction grating. The second order image is formed at an angle of 30 degrees from the central image. How many lines in a centimetre of the grating have been ruled?

Step 1 - Grating spacing from d sin(theta) = n lambda: d = n lambda / sin(theta) = 2 x 480 x 10^-9 / sin(30 degrees) = 960 x 10^-9 / 0.5 = 1.92 x 10^-6 m.

Step 2 - In centimetres, d = 1.92 x 10^-4 cm.

Step 3 - Lines per centimetre = 1 / d = 1 / (1.92 x 10^-4) = 5208, about 5.2 x 10^3 lines per cm.

Answer: 5.2 x 10^3 lines per cm (as printed in the book).

9.9 X-rays of wavelength 0.150 nm are observed to undergo a first order reflection at a Bragg angle of 13.3 degrees from a quartz (SiO2) crystal. What is the interplanar spacing of the reflecting planes in the crystal?

Step 1 - Bragg's law: 2 d sin(theta) = n lambda, with n = 1.

Step 2 - d = n lambda / (2 sin(theta)) = 0.150 / (2 x sin(13.3 degrees)).

Step 3 - sin(13.3 degrees) = 0.2300, so d = 0.150 / 0.460 = 0.326 nm.

Answer: 0.326 nm (as printed in the book).

9.10 An X-ray beam of wavelength lambda undergoes a first order reflection from a crystal when its angle of incidence to a crystal face is 26.5 degrees, and an X-ray beam of wavelength 0.097 nm undergoes a third order reflection when its angle of incidence to that face is 60.0 degrees. Assuming that the two beams reflect from the same family of planes, calculate (a) the interplanar spacing of the planes and (b) the wavelength lambda.

Step 1 - (a) For the second beam, Bragg's law gives 2 d sin(theta) = n lambda: 2 d sin(60.0 degrees) = 3 x 0.097 = 0.291 nm.

Step 2 - d = 0.291 / (2 x 0.8660) = 0.291 / 1.732 = 0.168 nm.

Step 3 - (b) For the first beam, 2 d sin(theta) = lambda, with n = 1: lambda = 2 x 0.168 x sin(26.5 degrees) = 0.336 x 0.4462 = 0.150 nm.

Answer: (a) 0.168 nm (b) 0.150 nm (as printed in the book).

Chapter 10 - Optical Instruments

Exercise 10 - Questions

10.1 What do you understand by linear magnification and angular magnification? Explain how a convex lens is used as a magnifier?

Linear magnification is the ratio of the size of the image to the size of the object, m = size of image / size of object = image distance / object distance. Angular magnification, or magnifying power, is the ratio of the angle subtended at the eye by the image formed with the instrument to the angle subtended at the unaided eye by the object placed at the least distance of distinct vision, 25 cm. A convex lens is used as a magnifier by placing the object within its focal length. The lens then forms an erect, virtual and magnified image on the same side as the object, at the near point or at infinity, and the eye sees the object under a larger angle than it would unaided.

10.2 Explain the difference between angular magnification and resolving power of an optical instrument. What limits the magnification of an optical instrument?

Angular magnification tells how much larger an object appears, that is, how many times bigger the angle subtended at the eye becomes. Resolving power tells how much detail can be seen, that is, the smallest separation between two close points that can still be distinguished as two separate points. The magnification of an optical instrument is limited by its resolving power: magnifying beyond it only enlarges a blurred image without revealing new detail, a condition called empty magnification. It is also limited by the aberrations of the lenses and by the loss of brightness as magnification increases.

10.3 Why would it be advantageous to use blue light with a compound microscope?

The resolving power of a microscope improves as the wavelength of the light used decreases, since the smallest distance that can be resolved is proportional to the wavelength. Blue light has a shorter wavelength than red light or the average wavelength of white light, so a microscope used with blue light can resolve finer detail and show smaller structures distinctly.

10.4 One can buy a cheap microscope for use by the children. The images seen in such a microscope have coloured edges. Why is this so?

Because of chromatic aberration. A cheap microscope uses simple, uncorrected lenses, and the refractive index of glass is different for different colours, so each colour is brought to a focus at a slightly different place. The image of a white object is therefore fringed with colours at its edges. Good microscopes use achromatic combinations of lenses that correct this defect.

10.5 Describe with the help of diagrams, how (a) a single biconvex lens can be used as a magnifying glass. (b) biconvex lenses can be arranged to form a microscope.

(a) A single biconvex lens as a magnifying glass: the object is placed within the focal length of the lens. A ray from the top of the object parallel to the axis is refracted through the far focus, and a ray through the centre of the lens passes undeviated. On the eye side these rays diverge, and their back-projections meet at an erect, virtual, magnified image on the same side of the lens as the object. (b) A microscope from two biconvex lenses: the objective, of very short focal length, is placed close to the object and forms a real, inverted, magnified intermediate image just inside the focal length of the eyepiece. The eyepiece then acts as a magnifying glass, forming a final virtual, magnified image at the least distance of distinct vision.

10.6 If a person was looking through a telescope at the full moon, how would the appearance of the moon be changed by covering half of the objective lens.

The appearance of the moon would not be changed in shape or size, only dimmed. Every part of the image is formed by light passing through the whole of the objective lens, so covering half of the objective only cuts off half the light. The full moon would still be seen complete, but fainter.

10.7 A magnifying glass gives a five times enlarged image at a distance of 25 cm from the lens. Find, by ray diagram, the focal length of the lens.

The magnifying power with the image at the near point is $M = 1 + d / f$, where $d = 25$ cm. So $5 = 1 + 25 / f$, giving $25 / f = 4$, and $f = 25 / 4 = 6.25$ cm. By the ray diagram, the object is placed 5 cm from the lens: with the virtual image at 25 cm,

the lens formula gives an object distance of 5.0 cm, and the image is five times as large as the object.

10.8 Identify the correct answer. (i) The resolving power of a compound microscope depends on; a. Length of the microscope. b. The diameter of the objective lens. c. The diameter of the eyepiece. d. The position of an observer's eye with regard to the eye lens. (ii) The resolving power of an astronomical telescope depends on: a. The focal length of the objective lens. b. The least distance of distinct vision of the observer. c. The focal length of the eye lens. d. The diameter of the objective lens.

(i) b. The diameter of the objective lens. The resolving power of a microscope depends on the aperture of its objective, which gathers the light and fixes the smallest distance that can be resolved; it does not depend on the length of the microscope, the eyepiece, or the position of the eye. (ii) d. The diameter of the objective lens. The resolving power of an astronomical telescope is proportional to the diameter of its objective, since a larger objective collects more light and resolves smaller angles.

10.9 Draw sketches showing the different light paths through a single-mode and a multi mode fibre. Why is the single-mode fibre preferred in telecommunications?

In a single-mode fibre the light travels along essentially one path, straight down the centre of the core, so all parts of a pulse arrive together. In a multi-mode fibre the light travels along many paths at different angles, bouncing off the core-cladding boundary, so the parts of a pulse that take steeper paths arrive later than those that take straighter paths. The single-mode fibre is preferred in telecommunications because it has no such modal dispersion: pulses stay sharp over long distances, so signals can be sent faster and with less distortion.

10.10 How the light signal is transmitted through the optical fibre?

Light is transmitted through an optical fibre by total internal reflection. A ray entering the core strikes the boundary between the core, of higher refractive index,

and the cladding, of lower refractive index, at an angle greater than the critical angle, and is reflected back into the core instead of escaping. It is reflected again and again along the fibre, zig-zagging its way to the far end with very little loss.

10.11 How the power is lost in optical fibre through dispersion? Explain.

Power is lost in an optical fibre through dispersion in two ways. In modal dispersion, different modes travel different path lengths and arrive at different times, so a pulse spreads out. In material (chromatic) dispersion, different wavelengths travel at different speeds in the glass, so a pulse containing several wavelengths also spreads out. As the pulse spreads, its peak power falls and neighbouring pulses begin to overlap, so the signal becomes weaker and harder to read. Absorption and scattering in the glass add further loss of power.

Exercise 10 - Numerical Problems

10.1 A converging lens of focal length 5.0 cm is used as a magnifying glass. If the near point of the observer is 25 cm and the lens is held close to the eye, calculate (i) the distance of the object from the lens (ii) the angular magnification. What is the angular magnification when the final image is formed at infinity?

Step 1 - The final image is at the near point, 25 cm from the lens, and virtual, so with the lens formula, 1 / f = 1 / v - 1 / u, with v = -25 cm and f = 5.0 cm: 1 / 5.0 = -1 / 25 - 1 / u.

Step 2 - So 1 / u = -1 / 25 - 1 / 5.0 = -(1 + 5) / 25 = -6 / 25, giving u = -4.17 cm. (i) The object is 4.2 cm from the lens.

Step 3 - (ii) Angular magnification with the image at the near point: M = 1 + d / f = 1 + 25 / 5.0 = 6.0.

Step 4 - With the final image at infinity, M = d / f = 25 / 5.0 = 5.0.

Answer: (i) 4.2 cm (ii) 6.0 ; 5.0 (as printed in the book).

10.2 A telescope objective has focal length 96 cm and diameter 12 cm. Calculate the focal length and minimum diameter of a simple eye piece lens for use with the telescope, if the linear magnification required is 24 times and all the light transmitted by the objective from a distant point on the telescope axis is to fall on the eye piece.

Step 1 - Magnifying power of the telescope M = f_objective / f_eyepiece = 24, so f_eyepiece = 96 / 24 = 4.0 cm.

Step 2 - For all the light from the objective to fall on the eyepiece, the diameter of the eyepiece must be at least the diameter of the beam, which is reduced by the magnification: diameter of eyepiece = diameter of objective / M = 12 / 24 = 0.50 cm.

Answer: f_e = 4.0 cm, dia = 0.50 cm (as printed in the book).

10.3 A telescope is made of an objective of focal length 20 cm and an eye piece of 5.0 cm, both convex lenses. Find the angular magnification.

Step 1 - Angular magnification of a telescope in normal adjustment: M = f_objective / f_eyepiece.

Step 2 - M = 20 / 5.0 = 4.0.

Answer: 4.0 (as printed in the book).

10.4 A simple astronomical telescope in normal adjustment has an objective of focal length 100 cm and an eye piece of focal length 5.0 cm. (i) Where is the final image formed? (ii) Calculate the angular magnification.

Step 1 - (i) In normal adjustment the intermediate image formed by the objective lies at the focal point of the eyepiece, so the eyepiece forms the final image at infinity, for relaxed viewing.

Step 2 - (ii) Angular magnification M = f_objective / f_eyepiece = 100 / 5.0 = 20.

Answer: (i) infinity (ii) 20 (as printed in the book).

10.5 A point object is placed on the axis of and 3.6 cm from a thin convex lens of focal length 3.0 cm. A second thin convex lens of focal length 16.0 cm is placed coaxial with the first and 26.0 cm from it on the side away from the object. Find the position of the final image produced by the two lenses.

Step 1 - Image by the first lens: 1 / v1 = 1 / f1 + 1 / u1 in magnitudes, 1 / v1 = 1 / 3.0 - 1 / 3.6 = 0.3333 - 0.2778 = 0.0556, so v1 = 18 cm. The first lens forms a real image 18 cm beyond it.

Step 2 - The second lens is 26.0 cm from the first, so this image is 26.0 - 18 = 8.0 cm in front of the second lens and acts as its object.

Step 3 - Image by the second lens: 1 / v2 = 1 / f2 - 1 / u2 = 1 / 16.0 - 1 / 8.0 = 0.0625 - 0.125 = -0.0625, so v2 = -16 cm.

Step 4 - The negative sign means the final image is virtual, 16 cm in front of the second lens, on the side from which the light comes.

Answer: 16 cm from second lens (as printed in the book).

10.6 A compound microscope has lenses of focal length 1.0 cm and 3.0 cm. An object is placed 1.2 cm from the object lens. If a virtual image is formed, 25 cm from the eye, calculate the separation of the lenses and the magnification of the instrument.

Step 1 - Image by the objective: 1 / v = 1 / f_o - 1 / u = 1 / 1.0 - 1 / 1.2 = 1 - 0.8333 = 0.1667, so v = 6.0 cm. The objective forms a real image 6.0 cm beyond it, with linear magnification 6.0 / 1.2 = 5.

Step 2 - The eyepiece forms the final virtual image at 25 cm: 1 / u_e = -1 / 25 - 1 / f_e = -0.04 - 0.3333 = -0.3733, so the intermediate image is 2.68 cm in front of the eyepiece.

Step 3 - Separation of the lenses = 6.0 + 2.68 = 8.68 cm, about 8.7 cm.

Step 4 - Magnifying power of the eyepiece = 1 + d / f_e = 1 + 25 / 3.0 = 9.33. Total magnification = 5 x 9.33 = 46.7, about 47.

Answer: 8.7 cm, 47 (as printed in the book).

10.7 Sodium light of wavelength 589 nm is used to view an object under a microscope. If the aperture of the objective is 0.90 cm, (i) find the limiting angle of resolution, (ii) using visible light of any wavelength, what is the maximum limit of resolution for this microscope.

Step 1 - (i) Limiting angle of resolution theta = 1.22 lambda / D = 1.22 x 589 x 10^-9 / 0.0090 = 7.99 x 10^-5 rad, about 8.0 x 10^-5 rad.

Step 2 - (ii) The best resolution comes from the shortest visible wavelength, violet light, about 400 nm: theta = 1.22 x 400 x 10^-9 / 0.0090 = 5.42 x 10^-5 rad, about 5.4 x 10^-5 rad.

Answer: (i) 8.0 x 10^-5 rad, (ii) 5.4 x 10^-5 rad (as printed in the book).

10.8 An astronomical telescope having magnifying power of 5 consist of two thin lenses 24 cm apart. Find the focal lengths of the lenses.

Step 1 - In normal adjustment the separation of the lenses equals the sum of the focal lengths: f_o + f_e = 24 cm. The magnifying power is M = f_o / f_e = 5, so f_o = 5 f_e.

Step 2 - Substituting: 5 f_e + f_e = 24, so 6 f_e = 24, f_e = 4 cm and f_o = 20 cm.

Answer: 20 cm, 4 cm (as printed in the book).

10.9 A glass light pipe in air will totally internally reflect a light ray if its angle of incidence is at least 39 degrees. What is the minimum angle for total internal reflection if pipe is in water? (Refractive Index of water = 1.33.)

Step 1 - In air, the critical angle is 39 degrees, so the refractive index of the glass is n = 1 / sin(39 degrees) = 1 / 0.6293 = 1.589.

Step 2 - In water, sin(critical angle) = n_water / n_glass = 1.33 / 1.589 = 0.8370.

Step 3 - Critical angle = sin^-1(0.8370) = 56.9 degrees, about 57 degrees.

Answer: 57 degrees (as printed in the book).

10.10 The refractive index of the core and cladding of an optical fibre are 1.6 and 1.4 respectively. Calculate (i) the critical angle for the interface (ii) the maximum angle of incidence in the air of a ray which enters the fibre and is incident at the critical angle on the interface.

Step 1 - (i) Critical angle at the core-cladding interface: sin(c) = n_cladding / n_core = 1.4 / 1.6 = 0.875, so c = sin^-1(0.875) = 61.0 degrees, about 61 degrees.

Step 2 - (ii) At the entrance face, the ray in the core makes an angle of (90 degrees - 61 degrees) = 29 degrees with the normal to the entrance face. By Snell's law at the entrance, sin(theta) = n_core x sin(29 degrees) = 1.6 x 0.4848 = 0.7757.

Step 3 - theta = sin^-1(0.7757) = 50.9 degrees, about 51 degrees.

Answer: (i) 61 degrees, (ii) 51 degrees (as printed in the book).

Chapter 11 - Heat and Thermodynamics

Exercise 11 - Questions

11.1 Why is the average velocity of the molecules in a gas zero but the average of the square of velocities is not zero?

The molecules of a gas move in all directions with equal probability, so their vector velocities cancel one another on average and the mean velocity is zero. The square of a velocity, however, is always positive, whether the molecule moves to the left or to the right, so the squares do not cancel, and the average of the squares, <v^2>, is not zero. Its square root is the root mean square speed of the molecules.

11.2 Why does the pressure of a gas in a car tyre increase when it is driven through some distance?

When a car is driven, the tyre is repeatedly flexed and friction acts between the tyre and the road, so work is done on the tyre and the air inside it, and the temperature of the air rises. The volume of the tyre is almost constant, and at constant volume the pressure of a gas is proportional to its absolute temperature, so the pressure in the tyre increases.

11.3 A system undergoes from state P1 V1 to state P2 V2 as shown in Fig. 11.12. What will be the change in internal energy?

Zero. The curve in Fig. 11.12 is marked 'Constant Temperature', so the change from state P1 V1 to state P2 V2 is an isothermal change: the pressure falls from 4 x 10^5 Pa to 1 x 10^5 Pa while the volume rises from 1 cm^3 to 4 cm^3, keeping P V constant. The internal energy of an ideal gas depends only on its temperature, so when the temperature does not change, the change in internal energy is zero.

11.4 Variation of volume by pressure is given in Fig 11.13. A gas is taken along the paths ABCDA, ABCA and A to A. What will be the change in internal energy?

Zero in every case. Internal energy is a state function: its change depends only on the initial and final states of the system, not on the path taken. In Fig. 11.13(a) the gas goes round the cycle ABCDA, in (b) along ABCA, and in (c) round the closed loop, so in each case it returns to its starting state A. The change in internal energy is therefore zero for all three paths, even though the work done and the heat exchanged differ from path to path.

11.5 Specific heat of a gas at constant pressure is greater than specific heat at constant volume. Why?

At constant volume, all the heat supplied goes into raising the internal energy, and hence the temperature, of the gas. At constant pressure, the gas expands as it is heated and does work against the outside pressure, so extra heat must be supplied

to do that work as well as to raise the temperature. The specific heat at constant pressure is therefore greater than at constant volume, and for one mole of an ideal gas, Cp - Cv = R.

11.6 Give an example of a process in which no heat is transferred to or from the system but the temperature of the system changes.

An adiabatic process, in which no heat is transferred to or from the system, but the temperature still changes. For example, when air is compressed rapidly in a bicycle pump, no appreciable heat escapes during the quick compression, yet the air and the pump become warm, because the work done on the air increases its internal energy. Conversely, a gas escaping rapidly from a spray can cools, because it does work at the expense of its internal energy.

11.7 Is it possible to convert internal energy into mechanical energy? Explain with an example.

Yes. A heat engine converts internal energy into mechanical energy: in a steam engine, heat from the boiler turns water into steam, and the expanding steam pushes a piston and does work. In a petrol engine, the hot gases produced by burning fuel expand and drive the piston. Only part of the heat can be converted into work; the rest must be rejected to a cold sink.

11.8 Is it possible to construct a heat engine that will not expel heat into the atmosphere?

No. By the second law of thermodynamics, no heat engine can convert all the heat it absorbs into work; some heat must always be rejected to a colder reservoir. An engine that expelled no heat into the atmosphere would be 100 percent efficient, which is impossible, and would be a perpetual motion machine of the second kind.

11.9 A thermos flask containing milk as a system is shaken rapidly. Does the temperature of milk rise?

Yes, very slightly. Shaking the flask does work on the milk, and since the thermos is

insulated, almost no heat escapes, so the work done increases the internal energy of the milk and its temperature rises a little. The rise is tiny, because the work done by shaking is small compared with the heat capacity of the milk.

11.10 What happens to the temperature of the room, when an airconditioner is left running on a table in the middle of the room?

The temperature of the room will rise, not fall. An air conditioner pumps heat from the cold side to the hot side, but it also consumes electrical energy, which ends up as heat as well. With both the cooling unit and the heat-rejecting unit inside the same room, all the heat taken from the room air is dumped back into the room, together with the heat from the motor, so the room as a whole warms up. An air conditioner cools a room only when its hot side rejects heat outside the room.

11.11 Can the mechanical energy be converted completely into heat energy? If so give an example.

Yes. Mechanical energy can be converted completely into heat energy. For example, when a moving car brakes to a stop, all of its kinetic energy is converted into heat in the brakes and tyres; rubbing the hands together converts the work done against friction into heat. The reverse is not possible: heat energy cannot be converted completely into mechanical energy.

11.12 Does entropy of a system increase or decrease due to friction?

It increases. Friction converts ordered mechanical energy into disordered thermal energy, that is, into random molecular motion. The disorder of the system and its surroundings therefore grows, and by the second law of thermodynamics the total entropy increases as a result of any natural process involving friction.

11.13 Give an example of a natural process that involves an increase in entropy.

Many natural processes involve an increase in entropy, for example the melting of ice at room temperature, the mixing of two gases, the flow of heat from a hot body to a cold body, the spreading of perfume through a room by diffusion, or the

dissolving of salt in water. In each case the system becomes more disordered and its entropy increases.

11.14 An adiabatic change is the one in which a. No heat is added to or taken out of a system b. No change of temperature takes place c. Boyle's law is applicable d. Pressure and volume remains constant

a. No heat is added to or taken out of a system. An adiabatic change is defined by zero heat transfer. Option (b), no change of temperature, describes an isothermal change; option (c), Boyle's law, also applies to isothermal changes; and option (d) is false, since pressure and volume both change in an adiabatic change.

11.15 Which one of the following process is irreversible? a. Slow compressions of an elastic spring b. Slow evaporation of a substance in an isolated vessel c. Slow compression of a gas d. A chemical explosion

d. A chemical explosion. An explosion is a fast, violent, dissipative process that cannot be made to retrace its steps. The other three processes are slow and quasi-static, a slow compression of a spring, a slow evaporation in an isolated vessel, and a slow compression of a gas, and can be reversed approximately, so they are nearly reversible.

11.16 An ideal reversible heat engine has a. 100% efficiency b. Highest efficiency c. An efficiency which depends on the nature of working substance d. None of these

b. Highest efficiency. An ideal reversible (Carnot) engine has the highest possible efficiency for working between two given temperatures, equal to 1 - T2 / T1, where T1 and T2 are the absolute temperatures of the source and sink. It is not 100 percent efficient, since T2 is never zero, and its efficiency does not depend on the nature of the working substance.

Exercise 11 - Numerical Problems

11.1 Estimate the average speed of nitrogen molecules in air under standard

conditions of pressure and temperature.

Step 1 - For a gas molecule, the mean square speed is given by <v^2> = 3 k T / m = 3 R T / M, where M is the molar mass.

Step 2 - For nitrogen, M = 28 g per mole = 0.028 kg per mole. At standard temperature T = 273 K and R = 8.31 J per mole per K.

Step 3 - Root mean square speed = sqrt(3 R T / M) = sqrt(3 x 8.31 x 273 / 0.028) = sqrt(6808 / 0.028) = sqrt(243,143) = 493 m s^-1.

Answer: 493 ms^-1 (as printed in the book; this is the root mean square speed, which the chapter uses as the average molecular speed).

11.2 Show that ratio of the root mean square speeds of molecules of two different gases at a certain temperature is equal to the square root of the inverse ratio of their masses.

Step 1 - The root mean square speed of a gas is v_rms = sqrt(3 k T / m), where k is Boltzmann's constant, T the absolute temperature and m the mass of one molecule.

Step 2 - For two gases at the same temperature T, v1 = sqrt(3 k T / m1) and v2 = sqrt(3 k T / m2).

Step 3 - Dividing: v1 / v2 = sqrt(3 k T / m1) / sqrt(3 k T / m2) = sqrt(m2 / m1).

Step 4 - The ratio of the root mean square speeds is therefore the square root of the inverse ratio of the molecular masses, as required.

11.3 A sample of gas is compressed to one half of its initial volume at constant pressure of 1.25 x 10^5 Nm^-2. During the compression, 100 J of work is done on the gas. Determine the final volume of the gas.

Step 1 - At constant pressure, the work done on the gas is W = P x (change in volume) = P x (V1 - V2).

Step 2 - The final volume is half the initial volume: V2 = V1 / 2, so the change in volume is V1 - V2 = V1 / 2.

Step 3 - So 100 = 1.25 x 10^5 x (V1 / 2), giving V1 / 2 = 100 / (1.25 x 10^5) = 8.0 x 10^-4 m^3.

Step 4 - The final volume is V2 = V1 / 2 = 8.0 x 10^-4 m^3.

Answer: 8 x 10^-4 m^3 (as printed in the book).

11.4 A thermodynamic system undergoes a process in which its internal energy decreases by 300 J. If at the same time 120 J of work is done on the system, find the heat lost by the system.

Step 1 - First law of thermodynamics: change in internal energy = heat supplied - work done by the system, delta U = Q - W.

Step 2 - Here delta U = -300 J (a decrease). Work done on the system is 120 J, so work done by the system is W = -120 J.

Step 3 - Substituting: -300 = Q - (-120) = Q + 120, so Q = -300 - 120 = -420 J.

Step 4 - The negative sign means heat is lost by the system: heat lost = 420 J.

Answer: -420 J (as printed in the book).

11.5 A carnot engine utilises an ideal gas. The source temperature is 227 degrees C and the sink temperature is 127 degrees C. Find the efficiency of the engine. Also find the heat input from the source and heat rejected to the sink when 10000 J of work is done.

Step 1 - Convert to absolute temperatures: source T1 = 227 + 273 = 500 K, sink T2 = 127 + 273 = 400 K.

Step 2 - Efficiency of a Carnot engine = 1 - T2 / T1 = 1 - 400 / 500 = 1 - 0.80 = 0.20 = 20%.

Step 3 - Efficiency = work done / heat input, so heat input Q1 = W / efficiency = 10000 / 0.20 = 50,000 J = 5.00 x 10^4 J.

Step 4 - Heat rejected to the sink Q2 = Q1 - W = 50,000 - 10,000 = 40,000 J = 4.00 x 10^4 J.

Answer: 20%, 5.00 x 10^4 J, 4.00 x 10^4 J (as printed in the book).

11.6 A reversible engine works between two temperatures whose difference is 100 degrees C. If it absorbs 746 J of heat from the source and rejects 546 J to the sink, calculate the temperature of the source and the sink.

Step 1 - For a reversible engine, Q1 / Q2 = T1 / T2, where Q1 = 746 J is absorbed from the source and Q2 = 546 J is rejected to the sink.

Step 2 - So T1 / T2 = 746 / 546 = 1.366. Also T1 - T2 = 100 K (a difference of 100 degrees C equals 100 K).

Step 3 - Substituting T1 = 1.366 T2: 1.366 T2 - T2 = 100, so 0.366 T2 = 100, giving T2 = 273 K.

Step 4 - T1 = 273 + 100 = 373 K. In Celsius: source 100 degrees C, sink 0 degrees C.

Answer: 100 degrees C, 0 degrees C (as printed in the book).

11.7 A mechanical engineer develops an engine, working between 327 degrees C and 27 degrees C and claims to have an efficiency of 52%. Does he claim correctly? Explain.

Step 1 - Convert to absolute temperatures: source T1 = 327 + 273 = 600 K, sink T2 = 27 + 273 = 300 K.

Step 2 - The maximum possible efficiency between these temperatures is the Carnot efficiency: 1 - T2 / T1 = 1 - 300 / 600 = 1 - 0.50 = 50%.

Step 3 - No real or ideal engine working between these temperatures can exceed 50% efficiency.

Step 4 - The claimed efficiency of 52% is greater than the maximum possible, so the claim is incorrect.

Answer: No (as printed in the book).

11.8 A heat engine performs 100 J of work and at the same time rejects 400 J of heat energy to the cold reservoirs. What is the efficiency of the engine?

Step 1 - Heat input from the source Q1 = work done + heat rejected = 100 + 400 = 500 J.

Step 2 - Efficiency = work done / heat input = 100 / 500 = 0.20 = 20%.

Answer: 20% (as printed in the book).

11.9 A Carnot engine whose low temperature reservoir is at 7 degrees C has an efficiency of 50%. It is desired to increase the efficiency to 70%. By how many degrees the temperature of the source be increased?

Step 1 - Sink temperature T2 = 7 + 273 = 280 K. With efficiency 50%: 0.50 = 1 - T2 / T1, so T1 = 280 / 0.50 = 560 K = 287 degrees C.

Step 2 - For efficiency 70%: 0.70 = 1 - 280 / T1', so T1' = 280 / 0.30 = 933 K = 660 degrees C.

Step 3 - Increase in source temperature = 660 - 287 = 373 degrees C (the same in kelvin).

Answer: 373 degrees C (as printed in the book).

11.10 A steam engine has a boiler that operates at 450 K. The heat changes water to steam, which drives the piston. The exhaust temperature of the outside air is about 300 K. What is maximum efficiency of this steam engine?

Step 1 - Maximum efficiency is the Carnot efficiency between the boiler temperature

and the exhaust temperature: efficiency = 1 - T2 / T1.

Step 2 - Efficiency = 1 - 300 / 450 = 1 - 0.667 = 0.333 = 33.3%.

Answer: 33% (as printed in the book).

11.11 336 J of energy is required to melt 1 g of ice at 0 degrees C. What is the change in entropy of 30 g of water at 0 degrees C as it is changed to ice at 0 degrees C by a refrigerator?

Step 1 - Heat released when 30 g of water freezes at 0 degrees C: Q = 30 x 336 = 10,080 J.

Step 2 - The temperature is 0 degrees C = 273 K. Entropy change = heat transferred / absolute temperature, with a negative sign because heat leaves the water: delta S = -Q / T = -10,080 / 273.

Step 3 - delta S = -36.9 J K^-1.

Answer: -36.8 J K^-1 (as printed in the book; the direct value is -36.9 J K^-1, the small difference being rounding of the temperature and latent heat).